

Homeostatic regulation of REM sleep by the preoptic area of the hypothalamus

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Abstract

Rapid-eye-movement sleep (REMs) is characterized by activated electroencephalogram (EEG) and muscle atonia, accompanied by vivid dreams. REMs is homeostatically regulated, ensuring that any loss of REMs is compensated by a subsequent increase in its amount. However, the neural mechanisms underlying the homeostatic control of REMs are largely unknown. Here, we show that GABAergic neurons in the preoptic area of the hypothalamus projecting to the tuberomammillary nucleus (POA^{GAD2} → TMN neurons) are crucial for the homeostatic regulation of REMs. POA^{GAD2} → TMN neurons are most active during REMs, and inhibiting them specifically decreases REMs. REMs restriction leads to an increased number and amplitude of calcium transients in POA^{GAD2} → TMN neurons, reflecting the accumulation of REMs pressure. Inhibiting POA^{GAD2} → TMN neurons during REMs restriction blocked the subsequent rebound of REMs. Our findings reveal a hypothalamic circuit whose activity mirrors the buildup of homeostatic REMs pressure during restriction and that is required for the ensuing rebound in REMs.

eLife assessment

This **valuable** study advances our understanding of the brain nuclei involved in rapid-eye movement (REM) sleep regulation. Using a combination of imaging, electrophysiology, and optogenetic tools, the study provides **convincing** evidence that inhibitory neurons in the preoptic area of the hypothalamus influence REM sleep. This work will be of interest to neurobiologists working on the brain circuits of sleep.

Introduction

REMs is homeostatically regulated as demonstrated in various species including mice, rats, cats and humans (Siegel and Gordon, 1965 ; Beersma et al., 1990 ; Benington et al., 1994 ; Endo et al., 1997 ; 1998 ; Rechtschaffen et al., 1999 ; Franken, 2002 ; Shea et al., 2008 ). REMs

restriction leads to an increased homeostatic need for REMs. During the subsequent recovery sleep, the amount of REMs is increased to compensate for the amount lost during restriction. While seminal dissection studies indicate a pontine origin of REMs, recent studies have revealed that neural populations in the hypothalamus, midbrain, amygdala and medulla regulate REMs by promoting or inhibiting REMs (Clément et al., 2011 [↗](#); Jegou et al., 2013 [↗](#); Hayashi et al., 2015 [↗](#); Van Dort et al., 2015 [↗](#); Weber et al., 2015 [↗](#), 2018 [↗](#); Chung et al., 2017 [↗](#); Torontali et al., 2019 [↗](#)). However, we do not have a clear understanding about the homeostatic mechanisms regulating REMs and which brain regions integrate homeostatic REMs pressure.

The preoptic area of the hypothalamus (POA) is crucial for sleep regulation. The POA contains neurons that become activated during sleep and that are sufficient and necessary for sleep (Von Economo, 1930 [↗](#); Nauta, 1946 [↗](#); McGinty and Sterman, 1968 [↗](#); Sallanon et al., 1989 [↗](#); Sherin et al., 1996 [↗](#); Szymusiak et al., 1998 [↗](#); Lu et al., 2000 [↗](#); Gong et al., 2004 [↗](#); Takahashi et al., 2009 [↗](#); Zhang et al., 2015 [↗](#); Kroeger et al., 2018 [↗](#)). Specifically, POA GABAergic neurons projecting to the tuberomammillary nucleus (POA^{GAD2} → TMN neurons) form a subpopulation of sleep-active POA neurons, and promote sleep when optogenetically activated (Chung et al., 2017 [↗](#)). A previous study demonstrated that c-Fos expression in the POA is increased in REMs restricted rats (Gvilia et al., 2006 [↗](#)). However, the molecular identity of POA neurons that become activated during heightened REMs pressure and whether their activity is necessary for homeostatic REMs regulation remains largely unknown.

In this study, using fiber photometry, we found that POA^{GAD2} → TMN neurons become gradually activated during non-rapid eye movement sleep (NREMs) before the onset of REMs and are most active during REMs. Optogenetic inhibition of POA^{GAD2} → TMN neurons significantly decreased REMs. We therefore hypothesized that the POA^{GAD2} → TMN neurons are well suited to encode homeostatic pressure for REMs. Using fiber photometry recordings combined with REMs restriction, we show that the activity of POA^{GAD2} → TMN neurons significantly increased during periods of heightened REMs pressure. Optogenetic inhibition of POA^{GAD2} → TMN neurons during REMs restriction prevented the subsequent increase of REMs during recovery sleep. Our findings identify a hypothalamic circuit regulating the homeostatic need for REMs.

Results

POA^{GAD2} → TMN neurons are most active during REMs

To monitor the population activity of POA^{GAD2} → TMN neurons *in vivo* during spontaneous sleep, we performed fiber photometry recordings. GAD2-Cre mice were injected with retrograde adeno-associated viruses (AAVs) encoding Cre-inducible GCaMP8s (AAVretro-FLEX-jGCaMP8s) into the TMN (Tervo et al., 2016 [↗](#); Zhang et al., 2023 [↗](#)), and an optic fiber was implanted into the POA (Fig. 1A [↗](#); virus expression and optic fiber tracts were located in the ventrolateral POA, lateral POA, and the lateral part of medial POA). The calcium activity of POA^{GAD2} → TMN neurons was significantly higher during REMs compared with that during wake and NREMs (Fig. 1B, C [↗](#); detailed statistical results are shown in Figure legends and Table S1).

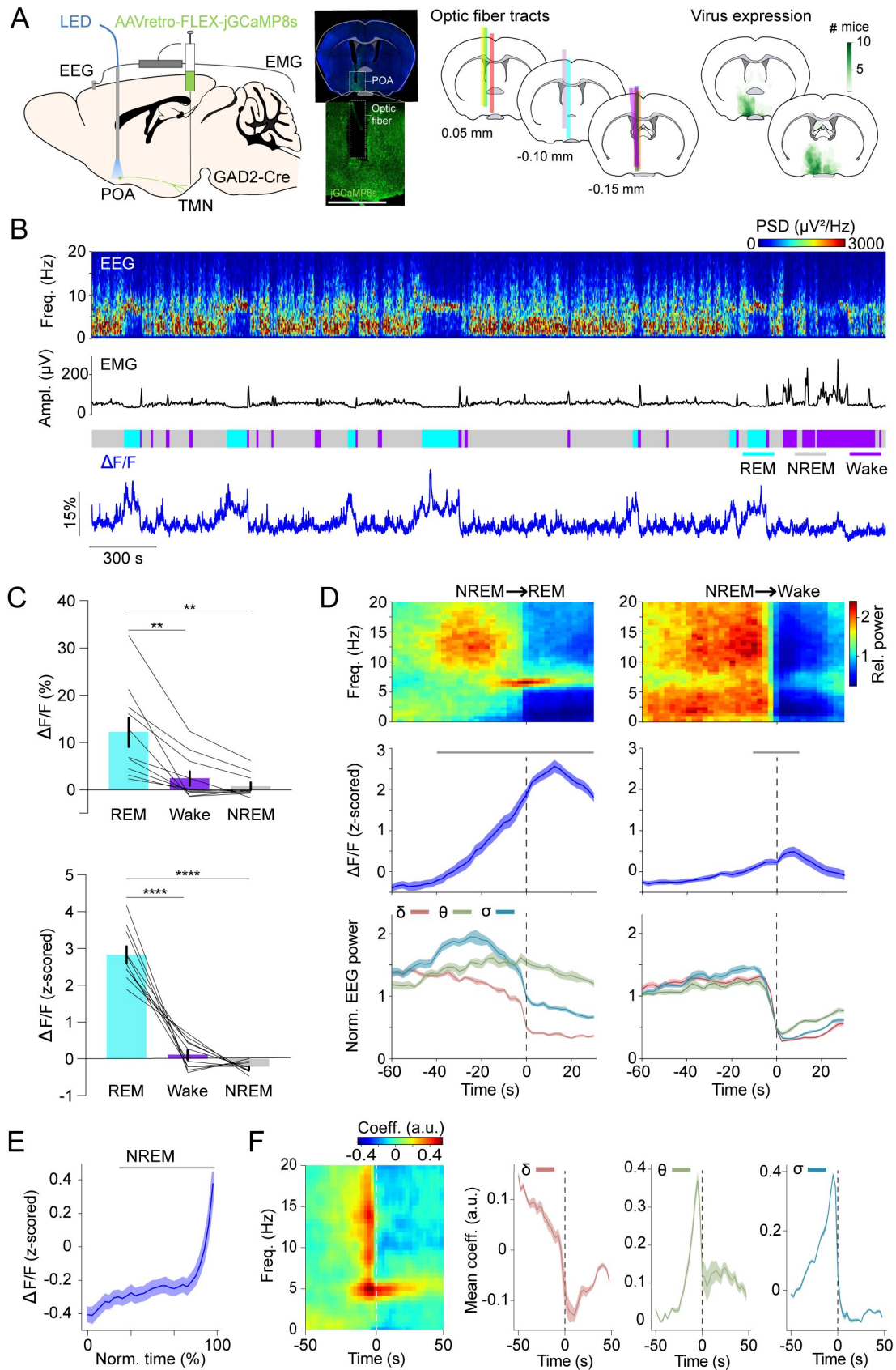


FIGURE 1

POA^{GAD2} → TMN neurons are most active during REMs.

(A) Left, schematic of fiber photometry with simultaneous EEG and EMG recordings. Mouse brain figure adapted from the Allen Reference Atlas - Mouse Brain (atlas.brain-map.org). Center left, fluorescence image of POA in a GAD2-Cre mouse injected with AAVretro-FLEX-jGCaMP8s into the TMN. Scale bar, 1 mm. Center right, location of fiber tracts. Each colored bar represents the location of optic fibers for photometry recordings. Right, heatmaps outlining areas with cell bodies expressing jGCaMP8. The green color code depicts how many mice the virus expression overlapped at the corresponding location (n = 10 mice).

(B) Example fiber photometry recording. Shown are EEG spectrogram, EMG amplitude, color-coded brain states, and $\Delta F/F$ signal.

(C) Non-normalized and z-scored $\Delta F/F$ activity during REMs, wake, and NREMs. Bars, averages across mice; lines, individual mice; error bars, $\pm$ s.e.m. One-way repeated measures (rm) ANOVA, $p = 9e-4$, $2e-6$ for non-normalized $\Delta F/F$ and z-scored $\Delta F/F$ signals; pairwise t-tests with Bonferroni correction, non-normalized $\Delta F/F$, $p = 0.0056$, 0.0039 for REMs vs. Wake and REMs vs. NREMs; z-scored $\Delta F/F$, $p = 6e-5$, $1e-7$. n = 10 mice.

(D) Average EEG spectrogram (top), z-scored $\Delta F/F$ activity (middle) and normalized EEG δ , θ and σ power (bottom) during NREMs → REMs transitions (left) and NREMs → Wake transitions (right). Shading, $\pm$ s.e.m. One-way rm ANOVA, $p = 3.18e-49$, $9.50e-8$ for NREMs → REMs and NREMs → Wake; pairwise t-tests with Holm-Bonferroni correction, NREMs → REMs $p < 0.0419$ between -40 s and 30 s, NREMs → Wake $p < 0.0106$ between -10 s and 10 s. Gray bar, period when $\Delta F/F$ activity was significantly different from baseline (-60 to -50 s). n = 10 mice.

(E) $\Delta F/F$ activity during NREMs. The duration of NREMs episodes was normalized in time, ranging from 0 to 100%. Shading, $\pm$ s.e.m. Pairwise t-tests with Holm-Bonferroni correction $p < 8.14e-9$ between 20 and 100. Gray bar, intervals where $\Delta F/F$ activity was significantly different from baseline (0 to 20%, the first time bin). n = 566 events.

(F) Left, linear filter mapping the normalized EEG spectrogram onto the POA^{GAD2} → TMN neural activity. Time point 0 s corresponds to the predicted neural activity. Right, coefficients of the linear filter for δ , θ and σ power band. Shading, $\pm$ s.e.m.

See **Table S1** for the actual p values.

We further analyzed the activity changes of the POA^{GAD2} → TMN neurons during NREMs → REMs or NREMs → wake transitions. We found that the calcium activity of POA^{GAD2} → TMN neurons during NREMs → REMs transitions becomes significantly increased 40 s before the REMs onset and remains elevated throughout REMs (**Fig. 1D**). During NREMs → wake transitions, the activity of POA^{GAD2} → TMN neurons started rising 10 s before the wake onset and remained elevated for 10 s after the onset (**Fig. 1D**). The $\Delta F/F$ activity gradually increased throughout NREMs episodes (**Fig. 1E**).

Given that both the θ and σ (6-9 and 10.5-16 Hz) power increased preceding the REMs onset (**Fig. 1D**), we investigated the relationship between the POA^{GAD2} → TMN neuron activity and the spectral composition of the EEG in more detail (**Fig. 1F**). We used a linear regression model to predict the current POA^{GAD2} → TMN neural activity based on the preceding and following spectral EEG features (Weber et al., 2010; Schott et al., 2023). We found that the activity of POA^{GAD2} → TMN neurons is preceded by an increase in the θ and σ power and reduction in the δ

(0.5–4.5 Hz) power (**Fig. 1F**). These changes in the EEG are characteristic for the stage of NREMs preceding REMs (Gottesmann, 1996) and their correlation with the activity of POA^{GAD2} → TMN neurons is consistent with a role of these neurons in promoting NREMs to REMs transitions.

In a complementary experiment, we monitored the calcium activity of POA^{GAD2} → TMN axonal fibers. GAD2-Cre mice were injected with AAV-FLEX-GCaMP6s into the POA and an optic fiber was implanted into the TMN (**Fig. S1A**). Similar to the activity found for POA^{GAD2} → TMN neurons using retrograde AAVs (**Fig. 1B, C**), POA^{GAD2} → TMN fibers were most active during REMs (**Fig. S1B, C**). During NREMs, the activity of POA^{GAD2} → TMN axonal fibers gradually increased before transitioning to REMs (**Fig. S1D**), demonstrating that the activity of POA^{GAD2} → TMN axonal fibers closely resembles that of cell bodies. In summary, these results demonstrate that the activity in POA^{GAD2} → TMN neurons increases prior to the onset of REMs episodes, following an increase in the θ and σ power in the EEG.

The TMN contains histamine neurons (TMN^{HIS}), and previous studies showed that POA^{GAD2} neurons innervate TMN^{HIS} neurons (Chung et al., 2017; Saito et al., 2018). Consistent with previous electrophysiological studies (Steininger et al., 1999; Vanni-Mercier et al., 2003; John et al., 2004; Takahashi et al., 2006), we found in photometry recordings that TMN^{HIS} neurons are highly active during wake and less active during NREMs and REMs (**Fig. S2A–C**). Consistent with this, the activity of TMN^{HIS} neurons became significantly activated after the transition from NREMs or REMs to wakefulness (**Fig. S2D**). The TMN^{HIS} neuron activity gradually decreased during NREMs episodes (**Fig. S2E**), possibly as a result of inhibitory inputs from POA^{GAD2} → TMN neurons (Chung et al., 2017). Moreover, examining the time course of TMN^{HIS} neurons between two successive REMs episodes (inter-REM interval), we found that their activity gradually decreases throughout the inter-REM interval, reaching its lowest level at the onset of REMs (**Fig. S2F**). This finding suggests that a minimal activity of TMN^{HIS} neurons is required for entering REMs, and suppression of TMN^{HIS} activity therefore likely facilitates transitions to REMs.

Inhibiting POA^{GAD2} → TMN neurons reduces REMs

To examine whether POA^{GAD2} → TMN neurons regulate REMs, we optogenetically inhibited these neurons using the bistable chloride channel SwiChR++ (Berndt et al., 2016; Smith et al., 2022; Stucynski et al., 2022). GAD2-Cre mice were bilaterally injected with retrograde AAVs encoding Cre-inducible SwiChR++ (AAVretro-DIO-SwiChR++-eYFP) or eYFP (AAVretro-DIO-eYFP) into the TMN followed by bilateral optic fiber implantation into the POA (**Fig. 2A**). We compared SwiChR++ and eYFP recordings with and without laser stimulation (473 nm, 2 s step pulses at 60 s intervals for 3 h, zeitgeber time [ZT] 2–5). SwiChR++-mediated inhibition of POA^{GAD2} → TMN neurons reduced the amount of REMs compared with recordings without laser stimulation in the same mice and eYFP mice with laser stimulation (**Fig. 2B, C**). The overall time spent in wake and NREMs was not altered by SwiChR++-mediated inhibition (**Fig. 2C, S3A, B**) suggesting that POA^{GAD2} → TMN neuron activity specifically regulates the amount of REMs.

Next, we compared the spectral composition of the EEG throughout recordings with or without laser stimulation in SwiChR++ and eYFP mice. During REMs, SwiChR++-mediated inhibition of POA^{GAD2} → TMN neurons increased the δ and σ power in the EEG compared with SwiChR++-without laser and eYFP-laser groups (**Fig. 2D**, **S3C**). The δ , θ , and σ power in the EEG during NREMs and wake was indistinguishable between SwiChR++ and eYFP groups (**Fig. S3D**).

Next, we investigated the effect of sustained inhibition of POA^{GAD2} → TMN neurons on the following 3 h recording without laser stimulation (**Fig. 2E**). Despite the reduction of REMs during the laser stimulation in SwiChR++ mice (**Fig. 2C**), there were no differences in the amount of REMs, NREMs, and wake between SwiChR++ and eYFP groups during the post laser recordings (**Fig. 2E, S3E, F**), suggesting that the loss of REMs during SwiChR++-mediated inhibition was not followed by a homeostatic increase in REMs.

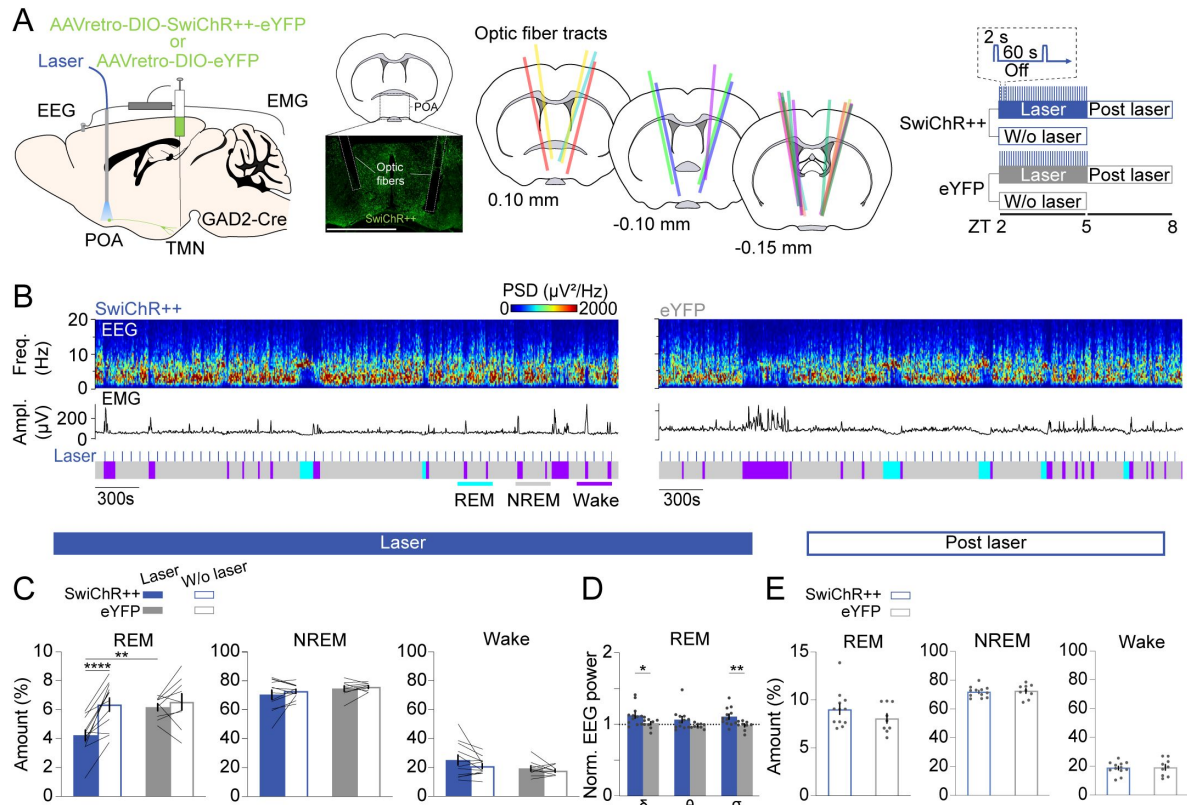


FIGURE 2

Inhibiting POA^{GAD2} → TMN neurons reduces REMs.

(A) Left, schematic of optogenetic inhibition experiments. Center left, fluorescence image of POA in a GAD2-CRE mouse injected with AAVretro-DIO-SwiChR++-eYFP into the TMN. Scale bar, 1 mm. Center right, location of optic fiber tracts. Each colored bar represents the location of an optic fiber. Right, experimental paradigm for laser stimulation (2 s step pulses at 60 s intervals) in SwiChR++ and eYFP-expressing mice.

(B) Example recording of a SwiChR++ (left) and eYFP mouse (right) with laser stimulation. Shown are EEG power spectra, EMG amplitude, and color-coded brain states.

(C) Percentage of time spent in REMs, NREMs and wakefulness with and without laser in SwiChR++ and eYFP mice. Mixed ANOVA, virus $p = 0.0629$, laser $p = 0.0003$, interaction $p = 0.0064$; t-tests with Bonferroni correction, SwiChR-laser vs. SwiChR-w/o laser $p = 3.00e-5$, SwiChR-laser vs. eYFP-laser $p = 0.0058$.

(D) Normalized EEG δ , θ , and σ power during REMs in SwiChR++ and eYFP mice with laser. Unpaired t-tests, SwiChR vs. eYFP $p = 0.0432$, 0.0099 for δ and σ .

(E) Percentage of time spent in REMs, NREMs, and wakefulness during post laser sessions.

(C) Bars, averages across mice; lines, individual mice; error bars, $\pm$ s.e.m.

(D, E) Bars, averages across mice; dots, individual mice; error bars, $\pm$ s.e.m. SwiChR++: $n = 12$ mice; eYFP: $n = 9$ mice

In a complementary experiment, we also investigated how inhibition of TMN^{HIS} neurons regulates REMs using the same inhibitory optogenetics protocol. HDC-Cre mice were bilaterally injected with AAVs encoding Cre-inducible SwiChR⁺⁺ (AAV₂-EF1a-DIO-SwiChR⁺⁺-eYFP) or eYFP (AAV₂-Ef1a-DIO-eYFP) into the TMN followed by bilateral optic fibers implantation into the TMN (**Fig. S4A**). Consistent with the previous experiment, we compared SwiChR⁺⁺ and eYFP recording with and without laser stimulation (**Fig. S4A, B**). SwiChR⁺⁺-mediated inhibition of TMN^{HIS} neurons increased the amount of REMs compared with recordings without laser stimulation in the same mice and eYFP mice with laser stimulation (**Fig. S4C**). Given that TMN also contains other types of neurons, inhibition of histamine neurons by POA^{GAD2} → TMN neurons may not be the sole source of the observed effect on REMs upon inhibition of POA^{GAD2} → TMN neurons.

Taken together, we first demonstrated that SwiChR⁺⁺-mediated inhibition of POA^{GAD2} → TMN neurons reduced the amount of REMs, supporting a necessary role of these neurons in REMs regulation. Second, the lost amount of REMs was not compensated for in the subsequent sleep suggesting that their activity is involved in the homeostatic regulation of REMs.

POA^{GAD2} → TMN neurons exhibit an increased number of calcium transients during REMs restriction

To probe whether the activity of POA^{GAD2} → TMN neurons changes during periods of high REMs pressure resulting from REMs restriction, we performed fiber photometry recordings combined with a closed-loop REMs restriction protocol (**Fig. 3A**). To detect the onset of REMs, we adapted a previously applied automatic REMs detection algorithm (Weber et al., 2015; 2018; Stucynski et al., 2022; Schott et al., 2023). Briefly, the animal's brain state was classified based on real-time analysis of the EEG/EMG signals. As soon as a REMs episode was detected, a small vibrating motor attached to the animal's head was turned on to terminate REMs by briefly awakening the animal (**Fig. 3A**, **S5A**) (Cardis et al., 2021; Osorio-Forero et al., 2023). Compared with the baseline recordings from the same mice during the same circadian time, we found that 6 h of REMs restriction (ZT 1.5-7.5) significantly reduced the amount of REMs and duration of REMs episodes, while increasing their frequency (**Fig. S5A, D-F**; Amount: Cohen's d [d] = -1.621; Duration: d = -2.669; Frequency: d = 1.294). The number of motor activations gradually increased as mice tried to enter REMs more frequently, indicating the accumulation of REMs pressure (**Fig. S5B**). Before the onset of the motor vibration, we found a clear increase in the EEG θ band reflecting NREMs to REMs transitions (**Fig. S5C**). We also investigated the effects of REMs restriction on the spectral composition of the EEG. We found an increased δ power for REMs during restriction (**Fig. S5G**). During rebound (ZT 7.5-8.5), the amount of REMs was significantly increased compared with that during the same circadian time due to an increased frequency of REMs episodes (**Fig. S5H-K**; Amount: d = 0.868; Frequency: d = 1.147), compensating for the amount of REMs lost during restriction.

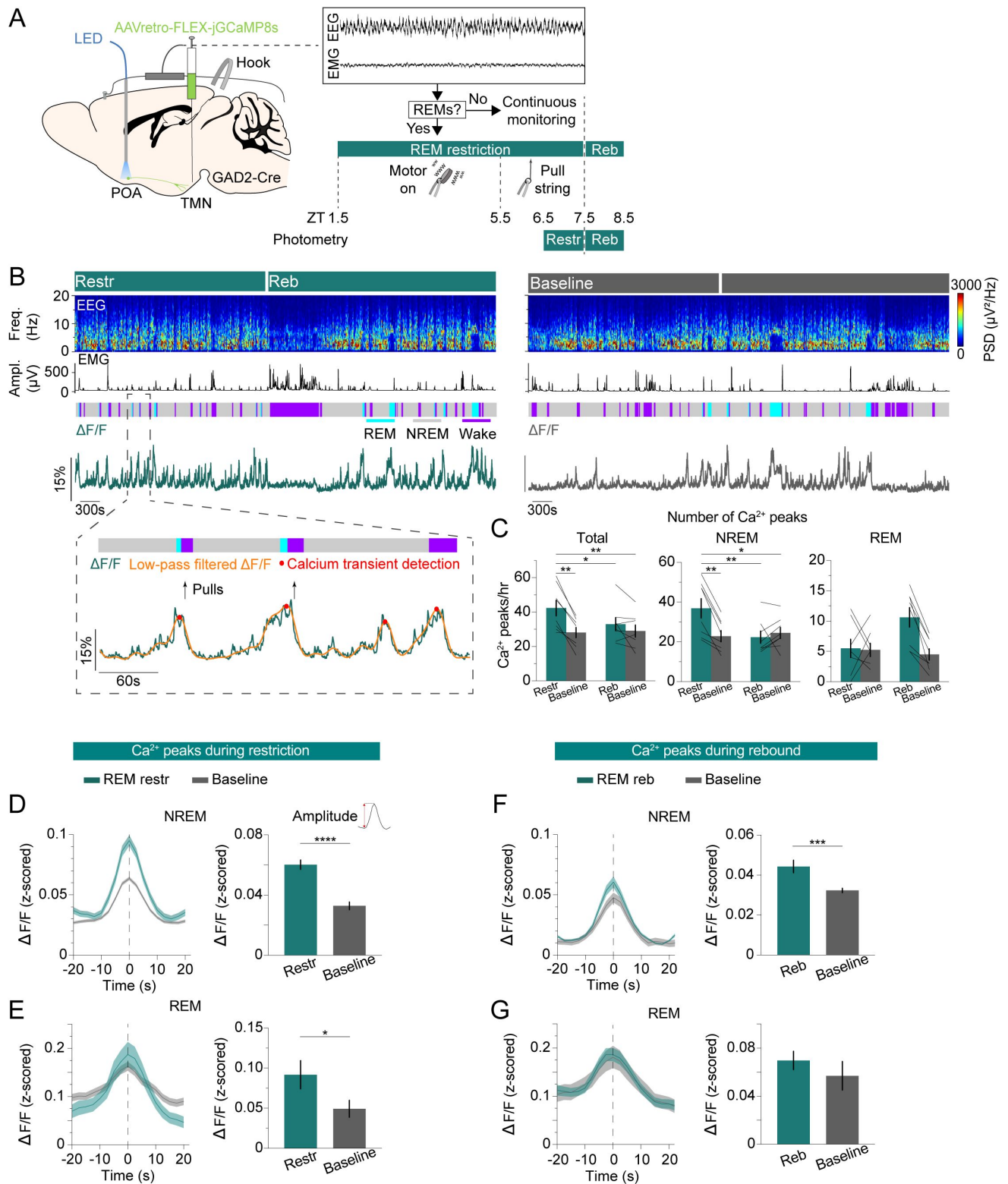


FIGURE 3

POA^{GAD2} → TMN neurons exhibit an increased number of calcium transients during REMs restriction.

(A) Schematic of REMs restriction/rebound and photometry recording experiments. The brain state was continuously monitored; once a REMs episode was detected, we used a vibrating motor (ZT 1.5-5.5) or pulled a string (ZT 5.5-7.5) attached to the mouse head to terminate REMs. Fiber photometry recordings were performed during REMs restriction (ZT 6.5-7.5) and rebound (ZT 7.5-8.5).

(B) Top, example fiber photometry recording. Shown are EEG spectrogram, EMG amplitude, color-coded brain states, and $\Delta F/F$ signal. Bottom, brain states, $\Delta F/F$ signal (green), low-pass filtered $\Delta F/F$ signal (orange), detected peaks (red) and pulls (arrows) during a selected interval (dashed box) at an expanded timescale.

(C) Number of calcium peaks during all states (left), NREMs (middle) and REMs (right). Bars, averages across mice; lines, individual mice; error bars, $\pm$ s.e.m. $n = 8$ mice.

Total: two-way rm ANOVA, treatment (baseline vs. manipulation) $p = 0.0031$, time $p = 0.0613$, interaction $p = 0.0189$; t-tests with Bonferroni correction, REM restriction (restr.) vs. baseline (ZT 6.5-7.5) $p = 0.0033$, restr. vs. REM rebound (reb.) $p = 0.0341$, restr. vs. baseline (ZT 7.5-8.5) $p = 0.0049$.

NREMs: two-way rm ANOVA, treatment $p = 0.0233$, time $p = 0.0363$, interaction $p = 0.003$; t-tests with Bonferroni correction, restr. vs. baseline (ZT 6.5-7.5) $p = 0.0057$, restr. vs. reb. $p = 0.0046$, restr. vs. baseline (ZT 7.5-8.5) $p = 0.0115$.

(D) Left, average NREMs calcium peaks during REMs restriction and baseline recordings. Right, average amplitude of the NREMs calcium peaks. The amplitude was calculated by subtracting the $\Delta F/F$ value 10 s before the peak from its value at the peak. Unpaired t-tests, $p = 6.30 \times 10^{-10}$. $n = 295$ and 196 peaks during restriction and baseline recordings.

(E) Left, average REMs calcium peaks during REMs restriction and baseline recordings. Right, average amplitude of the REMs calcium peaks. Unpaired t-tests, $p = 0.0437$. $n = 44$ and 42 peaks during restriction and baseline recordings.

(F) Left, average NREMs calcium peaks during REMs rebound and baseline recordings. Right, average amplitude of the NREMs calcium peaks. Unpaired t-tests, $p = 0.0002$. $n = 180$ and 196 peaks during rebound and baseline recordings.

(G) Left, average REMs calcium peaks during REMs rebound and baseline recordings. Right, average amplitude of the REMs calcium peaks. $n = 84$ and 36 peaks during rebound and baseline recordings.

(D-G) Bars, averages across trials; error bars, $\pm$ s.e.m; shadings, $\pm$ s.e.m.

During REMs restriction, we observed an increased number of calcium transients in the activity of POA^{GAD2} → TMN neurons as the REMs pressure increased (**Fig. 3A, B**). To detect and quantify these transients, we applied an algorithm previously applied to detect calcium events in fiber photometry recordings (Antila et al., 2022; Smith et al., 2022). Using fiber photometry, we monitored the population activity of POA^{GAD2} → TMN neurons during the last h of REMs restriction (ZT 6.5-7.5, referred as Restr.) and the first h of REMs rebound (ZT 7.5-8.5, referred as Reb.) and compared it with the activity during baseline recordings of the same mice on separate days at the same circadian time (**Fig. 3A, B**). To restrict REMs, we used a motor (ZT 1.5-5.5) as previously described or gently pulled a string attached to the animal's head (ZT 5.5-7.5) during the photometry recordings (**Fig. 3A**). The manual REMs deprivation was utilized during the last 2 h of REMs restriction to avoid potential motion artifacts from the vibrating motor that could contaminate calcium signals. We verified that REMs was adequately restricted (**Fig. S6**). During

restriction, the number of calcium transients of the POA^{GAD2} → TMN neurons was significantly increased compared with that during baseline recordings (**Fig. 3C**). In particular, the number of calcium peaks was significantly elevated during NREMs, likely reflecting an increased pressure to transition to REMs (**Fig. 3C**). As the duration of REMs episodes was largely reduced as a result of the restriction (**Fig. S6A**), the number of peaks during REMs was not changed.

We further examined the amplitude of calcium transients during both NREMs and REMs. During REMs restriction, the amplitude of NREMs and REMs calcium transients was significantly higher compared with that during the circadian baseline (**Fig. 3D, E**) and remained elevated during NREMs in the rebound phase (**Fig. 3F, G**). Overall, these findings show that during periods of high REMs pressure, the POA^{GAD2} → TMN neurons exhibit an increased number of calcium transients with higher amplitude compared with that during baseline levels suggesting that the activity of these neurons may reflect the heightened homeostatic need for REMs.

Inhibition of POA^{GAD2} → TMN neurons during REMs restriction reduces the REMs rebound

To investigate whether the activity of POA^{GAD2} → TMN neurons encodes REMs pressure and consequently facilitates the subsequent rebound in REMs, we optogenetically inhibited these neurons during the last 3 h of REMs restriction. GAD2-Cre mice were bilaterally injected with retrograde AAVs encoding Cre-inducible SwiChR++ (AAVretro-DIO-SwiChR++-eYFP) or eYFP (AAVretro-DIO-eYFP) into the TMN followed by bilateral optic fiber implantation into the POA (**Fig. 4A**). Mice underwent REMs restriction (6 h, ZT 1.5-7.5), and laser stimulation (2 s step pulses at 60 s intervals) was applied during the last 3 h (ZT 4.5-7.5), when the REMs pressure was highest (**Fig. S4B**). During restriction, SwiChR-mediated inhibition of POA^{GAD2} → TMN neurons decreased the percentage of REMs and reduced the frequency of REMs episodes with marginal significance (**Fig. 4B-D**, **S7A, B**, Amount: $d = -1.254$). We found that inhibition of POA^{GAD2} → TMN neurons during REMs restriction resulted in a reduced amount of REMs during the rebound compared with that in eYFP mice (**Fig. 4B, F, G, S7D**, **E**, Amount: $d = -1.317$). Thus, inactivating POA^{GAD2} → TMN neurons during heightened REMs pressure not only decreased the amount of REMs, but also prevented its homeostatic rebound during the following recovery sleep.

Finally, we examined the spectral composition of the EEG during the REMs restriction. We found that inhibition of POA^{GAD2} → TMN neurons during REMs restriction reduced the REMs δ power, NREMs θ and σ power compared with that in eYFP mice with laser stimulation (**Fig. 4E**, **S7C**). The reduced θ and σ power during NREMs could result from less attempts to enter REMs (**Fig. 4D**). During rebound, there were no significant differences in EEG δ , θ and σ power during REMs and NREMs (**Fig. 4H**, **S7F**).

Taken together, inhibiting POA^{GAD2} → TMN neurons during REMs restriction significantly decreased the amount of REMs and blocked the subsequent rebound in REMs. Our data suggest that the heightened activity of POA^{GAD2} → TMN neurons during sleep encodes the increased need for REMs and consequently plays an important role in the homeostatic response to REMs restriction.

Discussion

Our study demonstrates a role of POA^{GAD2} → TMN neurons in the homeostatic regulation of REMs. Using fiber photometry, we showed that the POA^{GAD2} → TMN neurons become activated throughout NREMs before transitioning to REMs, while being most active during REMs (**Fig. 1**). Sustained optogenetic inhibition of POA^{GAD2} → TMN neurons reduced the overall amount of REMs, and the loss of REMs was not compensated during the subsequent recovery sleep (**Fig. 2**).

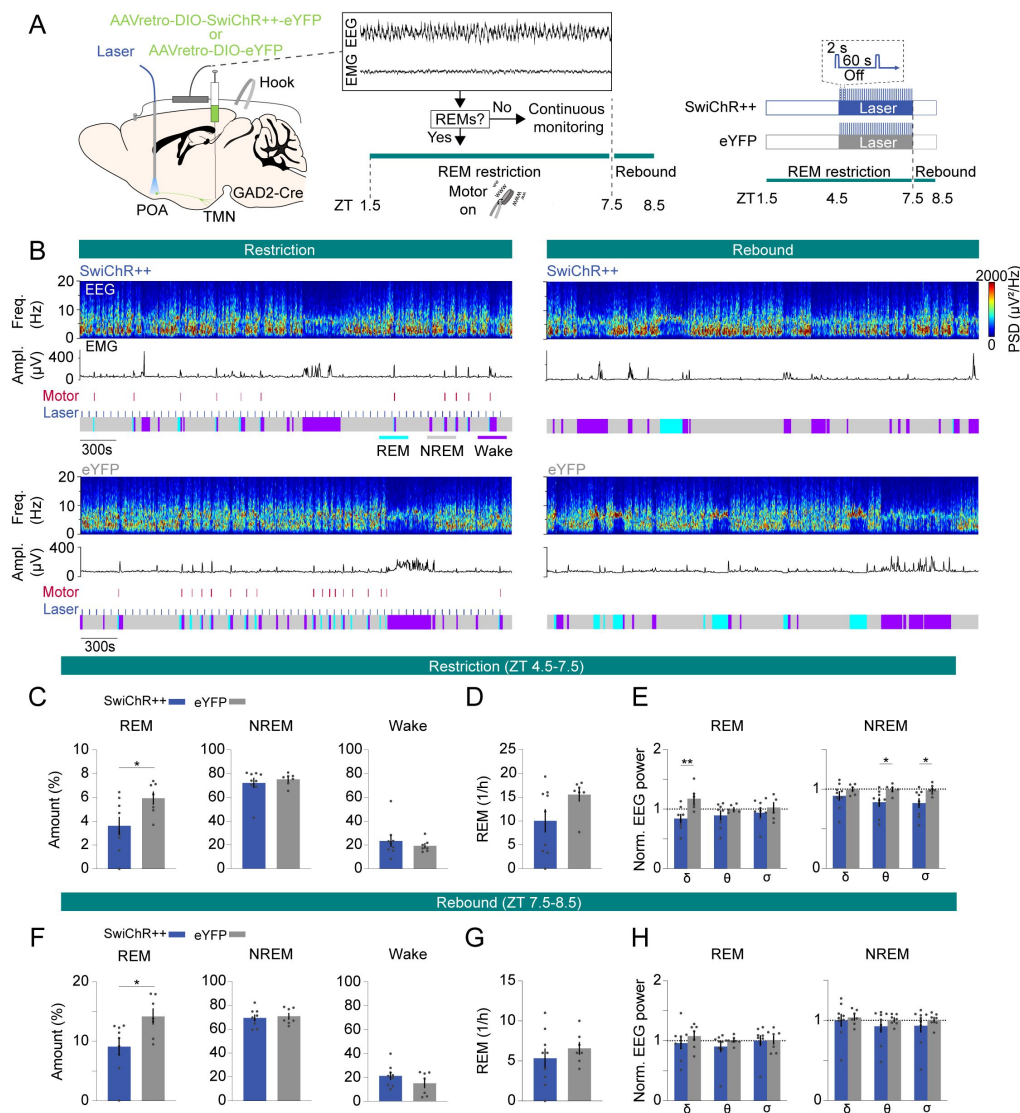


FIGURE 4

Inhibition of POA^{GAD2} → TMN neurons during REMs restriction attenuates the REMs rebound.

(A) Schematic of REMs restriction/rebound and optogenetic inhibition experiments. During closed-loop REMs restriction (ZT 1.5-7.5), a vibrating motor attached to the mouse head was used to terminate REMs. REMs was restricted for 6 h (ZT 1.5-7.5). During the last 3 h of restriction (ZT 4.5-7.5), laser stimulation (2 s step pulses at 60 s intervals) was applied in SwiChR++ and eYFP mice.

(B) Example sessions from a SwiChR++ (top) and eYFP mouse (bottom) during REMs restriction with laser stimulation (left) and rebound (right). Shown are EEG spectrogram, EMG amplitude, motor vibration events, laser and color-coded brain states.

(C) Percentage of time spent in REMs, NREMs and wakefulness during the last 3 h of REMs restriction with laser stimulation (ZT 4.5-7.5) in mice expressing SwiChR++ and eYFP. Unpaired t-tests, $p = 0.026$ for REMs amount.

(D) Frequency of REMs episodes during the last 3 h of REMs restriction with laser stimulation (ZT 4.5-7.5). Unpaired t-tests, $p = 0.0821$.

(E) Normalized EEG δ , θ and σ power during the last 3 h of REMs restriction with laser stimulation (ZT 4.5-7.5). Unpaired t-tests, $p = 0.0091$, 0.0332 , and 0.038 for REMs δ , NREMs θ and σ power.

(F) Percentage of time spent in REMs, NREMs and wakefulness during REMs rebound (ZT 7.5-8.5) in SwiChR++ and eYFP mice. Unpaired t-tests, $p = 0.0205$ for REMs amount.

(G) Frequency of REMs episodes during REMs rebound (ZT 7.5-8.5).

(H) Normalized EEG δ , θ and σ power during REMs rebound (ZT 7.5-8.5).

Bars, averages across mice; dots, individual mice; error bars, $\pm$ s.e.m. SwiChR++: $n = 9$ mice; eYFP: $n = 7$ mice

During the period of high REMs pressure, POA^{GAD2} → TMN neurons exhibited an increased number of calcium transients with elevated amplitude (**Fig. 3**). Optogenetic inhibition of POA^{GAD2} → TMN neurons during REMs restriction attenuated the subsequent rebound of REMs (**Fig. 4**). Our results suggest that the activity of POA^{GAD2} → TMN neurons reflects an increased need for REMs in the form of enhanced calcium transients and is required for the rebound following the loss of REMs.

The TMN contains histamine-producing neurons and antagonizing histamine signaling causes sleepiness (Watanabe et al., 1983; Panula et al., 1984; Lin et al., 1988; Bayliss et al., 1990; Haas et al., 2008; Uygun et al., 2016). Consistent with our photometry recordings (**Fig. S2**), electrophysiological recordings demonstrate that TMN^{HIS} neurons are most active during wakefulness and less active during NREMs and REMs (Steininger et al., 1999; Vanni-Mercier et al., 2003; John et al., 2004; Takahashi et al., 2006). Throughout NREMs, the activity of TMN^{HIS} neurons gradually decreased while that of POA^{GAD2} → TMN neurons showed an opposite pattern (**Fig. 1E, S2E**), which is likely in part the result of direct synaptic inputs from the POA^{GAD2} → TMN neurons to TMN^{HIS} neurons (Chung et al., 2017; Saito et al., 2018). TMN^{HIS} neurons in turn inhibit putative sleep-active POA neurons (Williams et al., 2014). Mutual inhibition between TMN^{HIS} and POA^{GAD2} → TMN neurons may explain their antagonistic activity pattern revealed in our fiber photometry recordings.

While many studies have focused on the role of the POA in regulating NREMs (Sherin et al., 1996; Szymusiak et al., 1998; Lu et al., 2000; Zhang et al., 2015; Kroeger et al., 2018; Ma et al., 2019), previous *in vivo* electrophysiological studies found that the majority of sleep-active neurons in the POA are most active during REMs (Osaka and Matsumura, 1995; Takahashi et al., 2009; Antila et al., 2022). Similarly, recent fiber photometry recordings demonstrated that GABAergic neurons in the POA and their subtypes expressing cholecystokinin, corticotropin-releasing hormone, tachykinin 1 or galanin are most active during REMs (Miracca et al., 2022; Smith et al., 2022). Deleting the NMDA receptor GluN1 subunit in the POA reduced REMs (Miracca et al., 2022), and in line with this, sustained optogenetic inhibition of POA^{GAD2} → TMN neurons specifically decreased REMs. Consistent with these studies, our findings also support an important role of the POA in REMs regulation and, in addition, provide evidence that these neurons are also part of the homeostat regulating REMs.

A previous study showed that the number of c-Fos positive POA neurons is positively correlated with the amount of REMs in rats (Lu et al., 2002). Moreover, REMs restriction led to an increase in the number of POA neurons expressing c-Fos, which was correlated with the number of attempts to enter REMs (Gvilia et al., 2006). Together with our findings that the number of calcium transients of POA^{GAD2} → TMN neurons increased during REMs restriction and that inhibition of these neurons blocked the following REMs rebound, these results support a crucial role of the POA in the homeostatic regulation of REMs. For the future, it would be interesting to test whether POA neurons projecting to other postsynaptic areas are also involved in the homeostatic regulation of REMs. A previous study showed that POA neurons projecting to REMs-regulatory pontine regions including the laterodorsal tegmental nucleus, locus coeruleus (LC) and dorsal raphe nucleus express increased levels of c-Fos after periods of dark exposure that increased REMs (Lu et al., 2002). However, the number of c-Fos positive POA neurons projecting to the LC was not increased upon REMs restriction, suggesting that this subpopulation may not be involved in the homeostatic regulation of REMs (Verret et al., 2006). Besides the TMN, the POA also projects to other REMs-regulatory regions such as the ventrolateral periaqueductal gray (vlPAG) and lateral hypothalamus (Steininger et al., 2001; Saito et al., 2018). Particularly, the projections to the vlPAG are of interest for future research, as GABAergic neurons in this area have been previously implicated in the homeostatic regulation of REMs (Hayashi et al., 2015; Weber et al., 2018). It remains to be tested whether POA^{GAD2} → TMN neurons also project to these brain regions to potentially regulate REMs homeostasis.

The cellular mechanisms underlying the elevated activity of POA neurons during high REMs pressure are unknown. Sleep-promoting neurons in the dorsal fan-shaped body of *Drosophila* display increased intrinsic neuronal excitability in response to sleep need (Donlea et al., 2014). REMs deprivation was shown to change the intrinsic excitability of hippocampal neurons and impact synaptic plasticity (McDermott et al., 2003; Mallick and Singh, 2011; Zhou et al., 2020). The elevated activity of POA^{GAD2} → TMN neurons during heightened REMs pressure may similarly be the result of an increased excitability. Given that the POA is also involved in the homeostatic regulation of NREMs (Sherin et al., 1996; Szymusiak et al., 1998; Gong et al., 2004; Zhang et al., 2015; Ma et al., 2019; Lu et al., 2000), it would be interesting to study how different POA subpopulations integrate the homeostatic need for NREMs and REMs.

Together, we have demonstrated a role of POA^{GAD2} → TMN neurons in the homeostatic regulation of REMs. REMs disturbances are observed in a variety of psychiatric disorders such as depression and PTSD and often precede their clinical onset (Ross et al., 1989; Gottesmann and Gottesman, 2007; Germain, 2013; Palagini et al., 2013). Elucidating the circuit mechanisms underlying the homeostatic regulation of REMs may provide novel therapeutic targets to specifically regulate and normalize REMs in these psychiatric disorders to alleviate associated symptoms and potentially slow down their progression.

Materials and methods

Mice

All experimental procedures were approved by the Institutional Animal Care and Use Committee (IACUC reference # 806197) at the University of Pennsylvania and conducted in compliance with the National Institutes of Health Office of Laboratory Animal Welfare Policy. Experiments were performed in male and female GAD2-IRES-Cre mice (#010802, Jackson Laboratory, generously donated by Taniguchi et al., 2011) or HDC-IRES-Cre mice (#021198, Jackson Laboratory, generously donated by Zecharia et al., 2012) aged 10-18 weeks, weighing 18-25 g at the time of surgery. Animals were group-housed with littermates on a 12h light/12h dark cycle (lights on 7 am and off 7 pm) with *ad libitum* access to food and water.

Viruses

Cre-dependent adeno-associated viral vectors were used to selectively express GCaMP, SwiChR++ or eYFP in POA^{GAD2} → TMN neurons, POA^{GAD2} → TMN projections or TMN^{HIS} neurons. pGP-AAV-Syn-FLEX-jGCaMP8s-WPRE was developed from the GENIE Project (Zhang et al., 2023) (162377-AAVrg, Addgene). pAAV-Syn-FLEX-GCaMP6s-WPRE-SV40 was developed from Douglas Kim & GENIE Project (Chen et al., 213) (Penn Vector Core or 100845-AAV1, Addgene). rAAV₂-Retro-Ef1α-DIO-SwiChR++-eYFP (R47730, UNC Vector Core). rAAV₂-Retro-Ef1α-DIO-eYFP (R49556, UNC Vector Core).

Surgical procedures

All procedures followed the IACUC guidelines for rodent survival surgery. Mice were anesthetized with isoflurane (1-2%) during the surgery, and placed on a stereotaxic frame (Kopf) while being on a heating pad to maintain body temperature. The skin was incised and small holes were drilled for virus injections and implantations of optic fibers and EEG/EMG electrodes.

For fiber photometry experiments to image POA^{GAD2} → TMN neurons (Fig. 1, 3), pGP-AAV-Syn-FLEX-jGCaMP8s-WPRE was injected (Nanoject II, Drummond Scientific) into the TMN (300 nl; AP -2.4 mm; ML -1 mm; DV -5.4 to -5.2 mm, relative to bregma) and an optic fiber (400 μm diameter) was implanted into the POA (AP 0.2 mm; ML -0.6 mm; DV -5.2 mm). For imaging the POA^{GAD2} → TMN axonal fibers (Fig. S1), pAAV-Syn-FLEX-GCaMP6s-WPRE-SV40 was injected into the POA

(300 nl) and an optic fiber was implanted into the TMN. To image TMN^{HIS} neurons (**Fig. S2**), pAAV-Syn-FLEX-GCaMP6s-WPRE-SV40 was injected into the TMN (300 nl) and an optic fiber was implanted into the TMN.

For optogenetic inhibition experiments (**Fig. 2**, [4](#)), rAAV₂-Retro-Ef1α-DIO-SwiChR++-eYFP (for inhibition group) or rAAV₂-Retro-Ef1α-DIO-eYFP (for control group) was bilaterally injected into the location on the reference picture the number of mice with overlapping virus expression, which was encoded using different green color intensities.

Polysomnographic recordings

All sleep recordings were performed in a cage to which the animal had been habituated for several days. All recordings were performed during the light phase between 8 am and 5 pm (ZT1-10) in sound-attenuating chambers. For sleep recordings, EEG and EMG signals were recorded using an RHD2132 amplifier (Intan Technologies, sampling rate 1 kHz) connected to an RHD USB interface board (Intan Technologies). For fiber photometry experiments, a calcium signal was recorded using an RZ5P amplifier (Tucker-Davis Technologies, sampling rate 1.5 kHz). EEG and EMG signals were referenced to a ground screw located on top of the cerebellum. At the start of each sleep recording, EEG and EMG electrodes were connected to flexible recording cables via small connectors. To determine the brain state of the animal, we first computed the EEG and EMG spectrogram for sliding, half-overlapping 5 s windows, resulting in 2.5 s time resolution. To estimate within each 5 s window the power spectral density (PSD), we performed Welch's method with Hanning window using sliding, half-overlapping 2 s intervals. Next, we computed the time-dependent δ (0.5 to 4 Hz), θ (5 to 12 Hz), σ (12 to 20 Hz), and high γ (100 to 150 Hz) power by integrating the EEG power in the corresponding ranges within the EEG spectrogram. In addition, we calculated the ratio of the θ and δ power (θ/δ) and the EMG power in the range of 50 to 500 Hz. For each power band, we used its temporal mean to separate it into a low and high part (except for the EMG and θ/δ ratio, where we used the mean plus one standard deviation as threshold). REMs was defined by a high θ/δ ratio, low EMG, and low δ power. NREMs was defined by high δ power, a low θ/δ ratio, and low EMG power. In addition, states with low EMG power, low δ power, but high σ power were scored as NREMs. Wake was defined by low δ power, high EMG power and high γ power (if not otherwise classified as REMs). Our automatic algorithm that has been previously used in (Weber et al., 2015 [4](#), 2018 [4](#); Chung et al., 2017 [4](#); Antila et al., 2022 [4](#); Smith et al., 2022; Stucynski et al., 2022 [4](#); Schott et al., 2023 [4](#)) has 90.256 % accuracy compared with the manual scoring by expert annotators. We manually verified the automatic classification using a graphical user interface visualizing the raw EEG and EMG signals, EEG spectrograms, EMG amplitudes, and the hypnogram to correct for errors, by visiting each single 2.5 s epoch in the hypnograms. The software for automatic brain state classification and manual scoring was programmed in Python (<https://github.com/tortugar/Lab/tree/master/PySleep> [4](#)).

Fiber photometry

Prior to the recording, the optic fiber and EEG/EMG electrodes were connected to flexible patch cables. For calcium imaging, a first LED (Doric lenses) generated the excitation wavelength of 465 nm and a second LED emitted 405 nm light, which served as control for bleaching and motion artifacts. 465 and 405 nm signals were modulated at two different frequencies, 210 and 330 Hz respectively. Both lights traveled through dichroic mirrors (Doric lenses) before entering a patch cable attached to the optic fiber. Fluorescence signals emitted by GCaMP8s or GCaMP6s were collected by the optic fiber and traveled via the patch cable through a dichroic mirror and GFP emission filter (Doric lenses) before entering a photoreceiver (Newport Co.). Photoreceiver signals were relayed to an RZ5P amplifier and demodulated into two signals using the Synapse software (Tucker-Davis Technologies), corresponding to the 465 and 405 nm excitation wavelengths. To analyze the calcium activity, we used custom-written Python scripts. First, both signals were low-pass filtered at 2 Hz using a 4th order digital Butterworth filter. Next, we fitted the 405 nm to the 465 nm signal using linear regression. Finally, the linear fit was subtracted from the 465 nm signal

to correct for photobleaching and/or motion artifacts, and the difference was divided by the linear fit yielding the $\Delta F/F$ signal. Both the fluorescence signals and EEG/EMG signals were simultaneously recorded using the RZ5P amplifier. Fiber photometry recordings were excluded if the signals suddenly shifted, likely due to a loose connection between the optic fiber and patch cable.

Optogenetic manipulation

Sleep recordings were performed during the light phase (ZT2-8). Mice were tethered to bilateral patch cables connected with the lasers and a flexible recording cable to record EEG/EMG signals. The recording started after 30 min of habituation. For optogenetic inhibition experiments, 2 s step pulses (1-3 mW, 60 s intervals) were generated by a blue laser (473 nm, Laserglow) and sent through the optic fiber (200 μ m diameter, Thorlabs) connected to the ferrule on the animal's head for 3 h (ZT2-5). This laser stimulation protocol was rationally designed based on previous reports of sustained inhibition and prior results that recapitulate similar findings as inhibitory chemogenetic techniques (Iyer et al., 2016 [DOI](#); Kim et al., 2016 [DOI](#); Wiegert et al., 2017 [DOI](#); Stucynski et al., 2022 [DOI](#)). TTL pulses to trigger the laser were controlled using a Raspberry Pi, which was controlled by a custom-programmed user interface programmed in Python. Following sustained inhibition, an additional 3 h (ZT5-8) recording was performed without laser stimulation (post-laser session). Baseline recordings were performed (without laser stimulation, ZT2-5), and counterbalanced across mice and days to avoid potential order effects. For each mouse, we collected 2-3 baseline and laser recordings each.

REMs restriction

We employed an automatic REMs detection algorithm (Weber et al., 2015 [DOI](#), 2018 [DOI](#); Stucynski et al., 2022 [DOI](#); Schott et al., 2023 [DOI](#)) and used a small vibrating motor to terminate/restrict REMs (Cardis et al., 2021 [DOI](#); Osorio-Forero et al., 2023 [DOI](#)) (<https://github.com/luthilab/IntanLuthiLab> [DOI](#)). A small vibrating motor (DC 3V Mini Vibration Motor, diameter: 10mm, thickness: 3mm, BOJACK) with a soldered lobster claw clasp was attached to a small wire hook secured in the dental cement of each mouse head. The motors had cables that were connected to a Raspberry Pi which controlled the motor onset and offset.

The automatic REMs detection algorithm determined whether the mouse was in REMs or not based on real-time spectral analysis of the EEG/EMG signals. The onset of REMs was defined as the time point where the EEG θ/δ ratio exceeded a threshold (mean + 1 std of θ/δ), which was calculated from the same mouse using previous recordings. As soon as a REMs episode was detected, the small vibrating motor turned on to terminate REMs and consequently woke up the mouse. The motor vibrated until the REMs episode was terminated, i.e. when the θ/δ ratio dropped below its mean value or if the EMG amplitude surpassed a threshold (mean + 0.5 std of amplitude). All REMs restriction experiments started at ZT1.5 and lasted until ZT7.5. Following the restriction, motors were turned off and mice were permitted to enter recovery sleep.

To monitor the calcium activity during REMs restriction using fiber photometry, we performed manual REMs restriction to avoid potential motion artifacts caused by the vibrating motor that could interfere with the integrity of the fiber photometry signals. Briefly mice underwent a 4 h automatic REMs restriction protocol as described above. During the last 2 h of restriction, REMs was detected manually based on the EEG and EMG signal by an experimenter who gently pulled on a string attached to the hook secured in the dental cement of the mouse head. To avoid potential photobleaching, the recording was performed during the last 1 h of REMs restriction and 1 h of REMs rebound. Each mouse also underwent a baseline recording on a separate day. The order of REMs restriction and baseline recordings were varied to minimize the impact of the experimental sequence on the results.

We also performed automatic REMs restriction combined with optogenetic inhibition. Each mouse underwent REMs restriction for 6 h, and we continuously delivered 2 s step pulses (1-3 mw, 473 nm, Laserglow) at 60 s intervals during the last 3 h of restriction (ZT4.5-7.5). Following REMs restriction, the recovery sleep was recorded.

Analysis of $\Delta F/F$ activity at brain state transitions and during NREMs

To calculate the neural activity changes relative to brain state transitions (Fig. 1D, Fig. S1D, Fig. S2D), we aligned the $\Delta F/F$ signals for transitions across all mice relative to the time point of the transition ($t = 0$ s). For each NREMs $\rightarrow$ REMs or NREMs $\rightarrow$ Wake transition, we ensured that the preceding NREMs episodes lasted for at least 60 s, only interrupted by short awakenings (≤ 20 s). To determine the time point at which the activity significantly started to increase or decrease, we used the first 10 s of NREMs as baseline. For REMs $\rightarrow$ Wake and Wake $\rightarrow$ NREMs transitions, the preceding REMs or Wake episode was at least 30 s long. Using one-way rm ANOVA, we tested whether the activity (downsampled to 10s bins) within each 10 s bin was significantly modulated throughout the transition (from -60 to 30 s). Finally, using pairwise t-tests with Holm-Bonferroni correction, we determined the time bins for which the activity significantly differed from the baseline bin (activity for bin -60 to -50 s). The time point for a given 10 s bin was set to its midpoint. To account for multiple comparisons, we divided the significance level ($\alpha = 0.05$) by the number of comparisons (Bonferroni correction). To analyze the activity throughout NREMs, we normalized the duration of all NREMs episodes and the corresponding $\Delta F/F$ signals to the same length (Fig. 1E, Fig. S2DE).

Spectrotemporal correlation analysis

To identify features of the EEG spectrogram associated with POA^{GAD2} $\rightarrow$ TMN calcium activity, we adapted a receptive field model (Weber et al., 2010; Schott et al., 2023) to predict POA^{GAD2} $\rightarrow$ TMN neural activity from the spectrogram. Intuitively, we estimated a spectrotemporal filter using linear regression that predicts for each time point the POA^{GAD2} $\rightarrow$ TMN calcium response. In more detail, we first computed the EEG spectrogram $E(f_i, t_j)$ using 2 s windows with 80 % overlap, resulting in a time resolution dt of 400 ms. Each spectrogram frequency was normalized by its mean power across the recording, and the parameter $E(f_i, t_j)$ specifies the relative amplitude of frequency f_i for time point t_j . We then downsampled the $\Delta F/F$ response using the same time resolution as for the spectrogram, and extracted all time bins with REMs, NREMs or wake for analysis.

To predict the calcium response, the EEG spectrogram was linearly filtered with the kernel $H(f_i, t_l)$. In analogy to receptive fields estimated using similar approaches for sensory neurons, we used the term “spectral field” for $H(f_i, t_l)$. The spectral field can be described as the set of coefficients that optimally relate the neural activity at time t_j to the EEG spectrogram at time t_{j+l} , where l represents the lag between the time points of the spectrogram and the neural response. The estimated frequency components of $H(f_i, t_l)$ ranged from $f_1 = 0.5$ Hz to $f_{nf} = 20$ Hz, while the time axis ranged from $-n_T * dt = -50$ s to $n_T * dt = 50$ s. Thus, $H(f_i, t_l)$ comprises n_f frequencies and $2 * n_T + 1$ time bins over a window from -50 to +50 seconds relative to the neural response. The convolution of the EEG spectrogram with the spectral field can be expressed as

$$\hat{r}(t_j) = r_0 + \sum_{l=-n_T}^{n_T} \sum_{i=1}^{n_f} E(f_i, t_{j+l}) H(f_i, t_l) \quad (\text{Eq. 1})$$

The scalar parameter r_0 denotes a constant offset, and $\hat{r}(t_j)$ denotes the predicted neural response. The optimal spectral field $H(f_i, t_l)$ minimizes the mean-squared error between the predicted and measured neural response at time t_j . To account for the large number of estimated parameters, we included a regularization term in the error function, which penalizes large kernel components

and therefore guards against overfitting of the model. The spectral field for each recording was estimated using 5-fold cross-validation to determine the regularization parameter (λ) that optimized the average model performance on the test sets. Kernels were averaged across all recordings for individual animals; the spectrotemporal correlation in **Fig. 1F** represents the mean spectral field across all mice.

Power spectral density and power estimation

The power spectral density (PSD) of the EEG was computed using Welch's method with the Hanning window for sliding, half-overlapping 2 s intervals. To calculate the power within a given frequency band, we approximated the corresponding area under the spectral density curve using a midpoint Riemann sum. To compute the EMG amplitude, we calculated the PSD of the EMG and integrated frequencies between 5 - 100 Hz. To test whether laser stimulation changed the spectral density during a specific brain state, we determined for each mouse the δ , θ or σ power for that state with and without laser. To compare the PSD between experimental and control mice, we calculated the δ , θ , and σ power with and without laser stimulation for each mouse and normalized the power values with laser by the power obtained for epochs without laser.

Detection of calcium transients

To detect calcium transients, we first filtered the $\Delta F/F$ signal with a zero-lag, 4th order digital Butterworth filter with cutoff frequency of 1/20 Hz. Next, prominent peaks in the signal were detected using the function `scipy.find_peaks` provided by the open source Python library `scipy` (<https://scipy.org>). As parameter for the peak prominence, we used 0.15 * distance between the 1st and 99th percentile of the distribution of the $\Delta F/F$ signal. A transient was defined as occurring during NREMs and REMs, if the peak overlapped with NREMs or REMs respectively. The calcium transient amplitude was calculated by using the values 10 s preceding the peak and subtracting that from the peak values at 0 s.

Statistical tests

Statistical analyses were performed using the python modules (`scipy.stats`, <https://scipy.org>; `pingouin`, <https://pingouin-stats.org>) and Prism v9.5.0.0 (GraphPad Software Inc). We did not predetermine sample sizes, but cohorts were similarly sized as in other relevant sleep studies ([Jego et al., 2013](#); [Ma et al., 2019](#)). All data collection was randomized and counterbalanced. All data are reported as mean $\pm$ s.e.m. A (corrected) p-value < 0.05 was considered statistically significant for all comparisons. Data were compared using unpaired t-tests, paired t-tests, one-way ANOVAs, or two-way ANOVAs followed by multiple comparisons as appropriate. The statistical results for the figures are presented in the Table S1, and Figure legends.

Data sharing plans

The code used for data analysis is publicly available under: <https://github.com/tortugar/Lab>. All the data are available from the corresponding author upon reasonable request.

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Author contributions

Conceptualization: J.M., F.W., and S.C.; Methodology: J.M., X.J., J.H., R.C., A.O-F. A.Lüthi, F.W., and S.C.; Software: J.M., X.J., and F.W.; Validation: J.M., A.Lin, and N.S.; Formal Analysis: J.M.; Investigation: J.M. and A.Lin; Resources: J.M., F.W., and S.C.; Data Curation: J.M. and A.Lin; Writing –Original Draft: J.M.; Writing –Review and Editing: J.M., R.C., A.O-F. A.Lüthi, F.W., and S.C.; Visualization: J.M.; Supervision: S.C.; Funding Acquisition: J.M. and S.C.

References

- Antila H., Kwak I., Choi A., Pisciotto A., Covarrubias I., Baik J., et al. (2022) **A noradrenergic-hypothalamic neural substrate for stress-induced sleep disturbances** *Proc. Natl. Acad. Sci. U. S. A* **119** <https://doi.org/10.1073/pnas.2123528119>
- Bayliss D. A., Wang Y. M., Zahnow C. A., Joseph D. R., Millhorn D. E (1990) **Localization of histidine decarboxylase mRNA in rat brain** *Mol. Cell. Neurosci* **1**:3–9 [https://doi.org/10.1016/1044-7431\(90\)90036-4](https://doi.org/10.1016/1044-7431(90)90036-4)
- Beersma D. G., Dijk D. J., Blok C. G., Everhardus I (1990) **REM sleep deprivation during 5 hours leads to an immediate REM sleep rebound and to suppression of non-REM sleep intensity** *Electroencephalogr. Clin. Neurophysiol* **76**:114–122 [https://doi.org/10.1016/0013-4694\(90\)90209-3](https://doi.org/10.1016/0013-4694(90)90209-3)
- Benington J. H., Woudenberg M. C., Heller H. C (1994) **REM-sleep propensity accumulates during 2-h REM-sleep deprivation in the rest period in rats** *Neurosci. Lett* **180**:76–80 [https://doi.org/10.1016/0304-3940\(94\)90917-2](https://doi.org/10.1016/0304-3940(94)90917-2)
- Berndt A., Lee S. Y., Wietek J., Ramakrishnan C., Steinberg E. E., Rashid A. J., et al. (2016) **Structural foundations of optogenetics: Determinants of channelrhodopsin ion selectivity** *Proc. Natl. Acad. Sci. U. S. A* **113**:822–829 <https://doi.org/10.1073/pnas.1523341113>
- Cardis R., Lecci S., Fernandez L. M., Osorio-Forero A., Chu Sin Chung P., Fulda S., et al. (2021) **Cortico-autonomic local arousals and heightened somatosensory arousability during NREMS of mice in neuropathic pain** *eLife* **10** <https://doi.org/10.7554/eLife.65835>
- Chen T.-W., Wardill T. J., Sun Y., Pulver S. R., Renninger S. L., Baohuan A., et al. (2013) **Ultrasensitive fluorescent proteins for imaging neuronal activity** *Nature* **499**:295–300 <https://doi.org/10.1038/nature12354>
- Chung S., Weber F., Zhong P., Tan C. L., Nguyen T. N., Beier K. T., et al. (2017) **Identification of preoptic sleep neurons using retrograde labelling and gene profiling** *Nature* **545**:477–481 <https://doi.org/10.1038/nature22350>
- Clément O., Sapin E., Béréd A., Fort P., Luppi P.-H (2011) **Evidence that neurons of the sublaterodorsal tegmental nucleus triggering paradoxical (REM) sleep are glutamatergic** *Sleep* **34**:419–423 <https://doi.org/10.1093/sleep/34.4.419>

Donlea J. M., Pimentel D., Miesenböck G (2014) **Neuronal Machinery of Sleep Homeostasis in *Drosophila*** *Neuron* **81**:860–872 <https://doi.org/10.1016/j.neuron.2013.12.013>

Endo T., Schwierin B., Borbély A. A., Tobler I (1997) **Selective and total sleep deprivation: effect on the sleep EEG in the rat** *Psychiatry Res* **66**:97–110 [https://doi.org/10.1016/S0165-1781\(96\)03029-6](https://doi.org/10.1016/S0165-1781(96)03029-6)

Endo T., Roth C., Landolt H. P., Werth E., Aeschbach D., Achermann P., et al. (1998) **Selective REM sleep deprivation in humans: effects on sleep and sleep EEG** *Am. J. Physiol* **274**:R1186–1194 <https://doi.org/10.1152/ajpregu.1998.274.4.R1186>

Franken P (2002) **Long-term vs. short-term processes regulating REM sleep** *J. Sleep Res* **11**:17–28 <https://doi.org/10.1046/j.1365-2869.2002.00275.x>

Germain A (2013) **Sleep disturbances as the hallmark of PTSD: where are we now?** *Am. J. Psychiatry* **170**:372–382 <https://doi.org/10.1176/appi.ajp.2012.12040432>

Gong H., McGinty D., Guzman-Marin R., Chew K.-T., Stewart D., Szymusiak R (2004) **Activation of c-fos in GABAergic neurones in the preoptic area during sleep and in response to sleep deprivation** *J. Physiol* **556**:935–946 <https://doi.org/10.1113/jphysiol.2003.056622>

Gottesmann C (1996) **The transition from slow-wave sleep to paradoxical sleep: evolving facts and concepts of the neurophysiological processes underlying the intermediate stage of sleep** *Neurosci. Biobehav. Rev* **20**:367–387 [https://doi.org/10.1016/0149-7634\(95\)00055-0](https://doi.org/10.1016/0149-7634(95)00055-0)

Gottesmann C., Gottesman I (2007) **The neurobiological characteristics of rapid eye movement (REM) sleep are candidate endophenotypes of depression, schizophrenia, mental retardation and dementia** *Prog. Neurobiol* **81**:237–250 <https://doi.org/10.1016/j.pneurobio.2007.01.004>

Gvilia I., Turner A., McGinty D., Szymusiak R (2006) **Preoptic Area Neurons and the Homeostatic Regulation of Rapid Eye Movement Sleep** *J. Neurosci* **26**:3037–3044 <https://doi.org/10.1523/JNEUROSCI.4827-05.2006>

Haas H. L., Sergeeva O. A., Selbach O (2008) **Histamine in the Nervous System** *Physiol. Rev* **88**:1183–1241 <https://doi.org/10.1152/physrev.00043.2007>

Hayashi Y., Kashiwagi M., Yasuda K., Ando R., Kanuka M., Sakai K., et al. (2015) **Cells of a common developmental origin regulate REM/non-REM sleep and wakefulness in mice** *Science* **350**:957–961 <https://doi.org/10.1126/science.aad1023>

Iyer S. M., Vesuna S., Ramakrishnan C., Huynh K., Young S., Berndt A., et al. (2016) **Optogenetic and chemogenetic strategies for sustained inhibition of pain** *Sci. Rep* **6** <https://doi.org/10.1038/srep30570>

Jego S., Glasgow S. D., Herrera C. G., Ekstrand M., Reed S. J., Boyce R., et al. (2013) **Optogenetic identification of a rapid eye movement sleep modulatory circuit in the hypothalamus** *Nat. Neurosci* **16**:1637–1643 <https://doi.org/10.1038/nn.3522>

John J., Wu M.-F., Boehmer L. N., Siegel J. M (2004) **Cataplexy-Active Neurons in the Hypothalamus: Implications for the Role of Histamine in Sleep and Waking Behavior** *Neuron* **42**:619–634 [https://doi.org/10.1016/S0896-6273\(04\)00247-8](https://doi.org/10.1016/S0896-6273(04)00247-8)

- Kim H., Ährlund-Richter S., Wang X., Deisseroth K., Carlén M (2016) **Prefrontal Parvalbumin Neurons in Control of Attention** *Cell* **164**:208–218 <https://doi.org/10.1016/j.cell.2015.11.038>
- Kroeger D., Absi G., Gagliardi C., Bandaru S. S., Madara J. C., Ferrari L. L., et al. (2018) **Galanin neurons in the ventrolateral preoptic area promote sleep and heat loss in mice** *Nat. Commun* **9** <https://doi.org/10.1038/s41467-018-06590-7>
- Lin J.-S., Sakai K., Jouvet M (1988) **Evidence for histaminergic arousal mechanisms in the hypothalamus of cat** *Neuropharmacology* **27**:111–122 [https://doi.org/10.1016/0028-3908\(88\)90159-1](https://doi.org/10.1016/0028-3908(88)90159-1)
- Lu J., Greco M. A., Shiromani P., Saper C. B (2000) **Effect of lesions of the ventrolateral preoptic nucleus on NREM and REM sleep** *J. Neurosci. Off. J. Soc. Neurosci* **20**:3830–3842 <https://doi.org/10.1523/JNEUROSCI.20-10-03830.2000>
- Lu J., Bjorkum A. A., Xu M., Gaus S. E., Shiromani P. J., Saper C. B (2002) **Selective activation of the extended ventrolateral preoptic nucleus during rapid eye movement sleep** *J. Neurosci. Off. J. Soc. Neurosci* **22**:4568–4576 <https://doi.org/10.1523/JNEUROSCI.22-11-04568.2002>
- Ma Y., Miracca G., Yu X., Harding E. C., Miao A., Yustos R., et al. (2019) **Galanin Neurons Unite Sleep Homeostasis and α 2-Adrenergic Sedation** *Curr. Biol. CB* **29**:3315–3322 <https://doi.org/10.1016/j.cub.2019.07.087>
- Mallick B. N., Singh A (2011) **REM sleep loss increases brain excitability: role of noradrenaline and its mechanism of action** *Sleep Med. Rev* **15**:165–178 <https://doi.org/10.1016/j.smrv.2010.11.001>
- McDermott C. M., LaHoste G. J., Chen C., Musto A., Bazan N. G., Magee J. C (2003) **Sleep deprivation causes behavioral, synaptic, and membrane excitability alterations in hippocampal neurons** *J. Neurosci. Off. J. Soc. Neurosci* **23**:9687–9695 <https://doi.org/10.1523/JNEUROSCI.23-29-09687.2003>
- McGinty D. J., Stermann M. B (1968) **Sleep suppression after basal forebrain lesions in the cat** *Science* **160**:1253–1255 <https://doi.org/10.1126/science.160.3833.1253>
- Miracca G., Anuncibay-Soto B., Tossell K., Yustos R., Vyssotski A. L., Franks N. P., et al. (2022) **NMDA Receptors in the Lateral Preoptic Hypothalamus Are Essential for Sustaining NREM and REM Sleep** *J. Neurosci. Off. J. Soc. Neurosci* **42**:5389–5409 <https://doi.org/10.1523/JNEUROSCI.0350-21.2022>
- Nauta W. J. H (1946) **Hypothalamic regulation of sleep in rats; an experimental study** *J. Neurophysiol* **9**:285–316 <https://doi.org/10.1152/jn.1946.9.4.285>
- Osaka T., Matsumura H (1995) **Noradrenaline inhibits preoptic sleep-active neurons through alpha 2-receptors in the rat** *Neurosci. Res* **21**:323–330 [https://doi.org/10.1016/0168-0102\(94\)00871-c](https://doi.org/10.1016/0168-0102(94)00871-c)
- Osorio-Forero A., Foustoukos G., Cardis R., Cherrad N., Devenoges C., Fernandez L. M. J., et al. (2023) **Locus coeruleus activity fluctuations set a non-reducible timeframe for mammalian NREM-REM sleep cycles** *bioRxiv* <https://doi.org/10.1101/2023.05.20.541586>
- Palagini L., Baglioni C., Ciapparelli A., Gemignani A., Riemann D (2013) **REM sleep dysregulation in depression: state of the art** *Sleep Med. Rev* **17**:377–390 <https://doi.org/10.1016/j.smrv.2012.11.001>

- Panula P., Yang H. Y., Costa E (1984) **Histamine-containing neurons in the rat hypothalamus** *Proc. Natl. Acad. Sci. U. S. A* **81**:2572–2576 <https://doi.org/10.1073/pnas.81.8.2572>
- Rechtschaffen A., Bergmann B. M., Gilliland M. A., Bauer K (1999) **Effects of method, duration, and sleep stage on rebounds from sleep deprivation in the rat** *Sleep* **22**:11–31 <https://doi.org/10.1093/sleep/22.1.11>
- Ross R. J., Ball W. A., Sullivan K. A., Caroff S. N (1989) **Sleep disturbance as the hallmark of posttraumatic stress disorder** *Am. J. Psychiatry* **146**:697–707 <https://doi.org/10.1176/ajp.146.6.697>
- Saito Y. C., Maejima T., Nishitani M., Hasegawa E., Yanagawa Y., Mieda M., et al. (2018) **Monoamines Inhibit GABAergic Neurons in Ventrolateral Preoptic Area That Make Direct Synaptic Connections to Hypothalamic Arousal Neurons** *J. Neurosci. Off. J. Soc. Neurosci* **38**:6366–6378 <https://doi.org/10.1523/JNEUROSCI.2835-17.2018>
- Sallanon M., Denoyer M., Kitahama K., Aubert C., Gay N., Jouvet M (1989) **Long-lasting insomnia induced by preoptic neuron lesions and its transient reversal by muscimol injection into the posterior hypothalamus in the cat** *Neuroscience* **32**:669–683 [https://doi.org/10.1016/0306-4522\(89\)90289-3](https://doi.org/10.1016/0306-4522(89)90289-3)
- Schott A. L., Baik J., Chung S., Weber F (2023) **A medullary hub for controlling REM sleep and pontine waves** *Nat. Commun* **14** <https://doi.org/10.1038/s41467-023-39496-0>
- Shea J. L., Mochizuki T., Sagvaag V., Aspevik T., Bjorkum A. A., Datta S (2008) **Rapid eye movement (REM) sleep homeostatic regulatory processes in the rat: changes in the sleep-wake stages and electroencephalographic power spectra** *Brain Res* **1213**:48–56 <https://doi.org/10.1016/j.brainres.2008.03.062>
- Sherin J. E., Shiromani P. J., McCarley R. W., Saper C. B (1996) **Activation of Ventrolateral Preoptic Neurons During Sleep** *Science* **271**:216–219 <https://doi.org/10.1126/science.271.5246.216>
- Siegel J., Gordon T. P (1965) **PARADOXICAL SLEEP: DEPRIVATION IN THE CAT** *Science* **148**:978–980 <https://doi.org/10.1126/science.148.3672.978>
- Smith J., Honig-Frand A., Antila H., Choi A., Kim H., Beier K. T., et al. (2023) **Regulation of stress-induced sleep fragmentation by preoptic glutamatergic neurons** *Curr. Biol* **0** <https://doi.org/10.1016/j.cub.2023.11.035>
- Steininger T. L., Alam Md. N., Gong H., Szymusiak R., McGinty D. (1999) **Sleep-waking discharge of neurons in the posterior lateral hypothalamus of the albino rat** *Brain Res* **840**:138–147 [https://doi.org/10.1016/S0006-8993\(99\)01648-0](https://doi.org/10.1016/S0006-8993(99)01648-0)
- Steininger T. L., Gong H., McGinty D., Szymusiak R (2001) **Subregional organization of preoptic area/anterior hypothalamic projections to arousal-related monoaminergic cell groups** *J. Comp. Neurol* **429**:638–653
- Stucynski J. A., Schott A. L., Baik J., Chung S., Weber F (2022) **Regulation of REM sleep by inhibitory neurons in the dorsomedial medulla** *Curr. Biol* **32**:37–50 <https://doi.org/10.1016/j.cub.2021.10.030>

- Szymusiak R., Alam N., Steininger T. L., McGinty D (1998) **Sleep-waking discharge patterns of ventrolateral preoptic/anterior hypothalamic neurons in rats** *Brain Res* **803**:178–188 [https://doi.org/10.1016/s0006-8993\(98\)00631-3](https://doi.org/10.1016/s0006-8993(98)00631-3)
- Takahashi K., Lin J.-S., Sakai K (2006) **Neuronal Activity of Histaminergic Tuberomammillary Neurons During Wake–Sleep States in the Mouse** *J. Neurosci* **26**:10292–10298 <https://doi.org/10.1523/JNEUROSCI.2341-06.2006>
- Takahashi K., Lin J.-S., Sakai K (2009) **Characterization and mapping of sleep-waking specific neurons in the basal forebrain and preoptic hypothalamus in mice** *Neuroscience* **161**:269–292 <https://doi.org/10.1016/j.neuroscience.2009.02.075>
- Taniguchi H., He M., Wu P., Kim S., Paik R., Sugino K., et al. (2011) **A resource of Cre driver lines for genetic targeting of GABAergic neurons in cerebral cortex** *Neuron* **71**:995–1013 <https://doi.org/10.1016/j.neuron.2011.07.026>
- Tervo D. G. R., Hwang B.-Y., Viswanathan S., Gaj T., Lavzin M., Ritola K. D., et al. (2016) **A Designer AAV Variant Permits Efficient Retrograde Access to Projection Neurons** *Neuron* **92**:372–382 <https://doi.org/10.1016/j.neuron.2016.09.021>
- Torontali Z. A., Fraigne J. J., Sanghera P., Horner R., Peever J (2019) **The Sublaterodorsal Tegmental Nucleus Functions to Couple Brain State and Motor Activity during REM Sleep and Wakefulness** *Curr. Biol. CB* **29**:3803–3813 <https://doi.org/10.1016/j.cub.2019.09.026>
- Uygun D. S., Ye Z., Zecharia A. Y., Harding E. C., Yu X., Yustos R., et al. (2016) **Bottom-Up versus Top-Down Induction of Sleep by Zolpidem Acting on Histaminergic and Neocortex Neurons** *J. Neurosci* **36**:11171–11184 <https://doi.org/10.1523/JNEUROSCI.3714-15.2016>
- Van Dort C. J., Zachs D. P., Kenny J. D., Zheng S., Goldblum R. R., Gelwan N. A., et al. (2015) **Optogenetic activation of cholinergic neurons in the PPT or LDT induces REM sleep** *Proc. Natl. Acad. Sci. U. S. A* **112**:584–589 <https://doi.org/10.1073/pnas.1423136112>
- Vanni-Mercier G., Gigout S., Debilly G., Lin J. S (2003) **Waking selective neurons in the posterior hypothalamus and their response to histamine H3-receptor ligands: an electrophysiological study in freely moving cats** *Behav. Brain Res* **144**:227–241 [https://doi.org/10.1016/s0166-4328\(03\)00091-3](https://doi.org/10.1016/s0166-4328(03)00091-3)
- Verret L., Fort P., Gervasoni D., Léger L., Luppi P.-H (2006) **Localization of the neurons active during paradoxical (REM) sleep and projecting to the locus coeruleus noradrenergic neurons in the rat** *J. Comp. Neurol* **495**:573–586 <https://doi.org/10.1002/cne.20891>
- Von Economo C. (1930) **Sleep as a problem of localization** *J. Nerv. Ment. Dis* **71**:249–259 <https://doi.org/10.1097/00005053-193003000-00007>
- Watanabe T., Taguchi Y., Hayashi H., Tanaka J., Shiosaka S., Tohyama M., et al. (1983) **Evidence for the presence of a histaminergic neuron system in the rat brain: an immunohistochemical analysis** *Neurosci. Lett* **39**:249–254 [https://doi.org/10.1016/0304-3940\(83\)90308-7](https://doi.org/10.1016/0304-3940(83)90308-7)
- Weber F., Machens C. K., Borst A (2010) **Spatiotemporal response properties of optic-flow processing neurons** *Neuron* **67**:629–642 <https://doi.org/10.1016/j.neuron.2010.07.017>
- Weber F., Chung S., Beier K. T., Xu M., Luo L., Dan Y (2015) **Control of REM sleep by ventral medulla GABAergic neurons** *Nature* **526**:435–438 <https://doi.org/10.1038/nature14979>

- Weber F., Hoang Do J. P., Chung S., Beier K. T., Bikov M., Saffari Doost M., et al. (2018) **Regulation of REM and Non-REM Sleep by Periaqueductal GABAergic Neurons** *Nat. Commun* **9** <https://doi.org/10.1038/s41467-017-02765-w>
- Wiegert J. S., Mahn M., Prigge M., Printz Y., Yizhar O (2017) **Silencing Neurons: Tools, Applications, and Experimental Constraints** *Neuron* **95**:504–529 <https://doi.org/10.1016/j.neuron.2017.06.050>
- Williams R. H., Chee M. J. S., Kroeger D., Ferrari L. L., Maratos-Flier E., Scammell T. E., et al. (2014) **Optogenetic-mediated release of histamine reveals distal and autoregulatory mechanisms for controlling arousal** *J. Neurosci. Off. J. Soc. Neurosci* **34**:6023–6029 <https://doi.org/10.1523/JNEUROSCI.4838-13.2014>
- Zecharia A. Y., Yu X., Götz T., Ye Z., Carr D. R., Wulff P., et al. (2012) **GABAergic Inhibition of Histaminergic Neurons Regulates Active Waking But Not the Sleep–Wake Switch or Propofol-Induced Loss of Consciousness** *J. Neurosci* **32**:13062–13075 <https://doi.org/10.1523/JNEUROSCI.2931-12.2012>
- Zhang Y., Rózsa M., Liang Y., Bushey D., Wei Z., Zheng J., et al. (2023) **Fast and sensitive GCaMP calcium indicators for imaging neural populations** *Nature* **615**:884–891 <https://doi.org/10.1038/s41586-023-05828-9>
- Zhang Z., Ferretti V., Güntan İ., Moro A., Steinberg E. A., Ye Z., et al. (2015) **Neuronal ensembles sufficient for recovery sleep and the sedative actions of $\alpha 2$ adrenergic agonists** *Nat. Neurosci* **18**:553–561 <https://doi.org/10.1038/nn.395>
- Zhou Y., Lai C. S. W., Bai Y., Li W., Zhao R., Yang G., et al. (2020) **REM sleep promotes experience-dependent dendritic spine elimination in the mouse cortex** *Nat. Commun* **11** <https://doi.org/10.1038/s41467-020-18592-5>

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Editors

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Reviewer #1 (Public Review):

This paper identifies GABA cells in the preoptic hypothalamus and others in the posterior hypothalamus which are involved in REM sleep rebound (the increase in REM sleep) after selective REM sleep deprivation. By calcium photometry, these preoptic cells are most active during REM, and show more calcium signals during REM deprivation, suggesting they respond to "REM pressure". Inhibiting these cells ontogenetically diminishes REM sleep. The optogenetic and photometry work is carried out to a high standard, the paper is well written,

and the findings are interesting and enhance our understanding of REM sleep regulation. The new findings make it clear that as for the circuitry that regulates NREM sleep, REM sleep circuitry is also quite distributed in the brain. It is unclear if there is a true "REM center". The study of mechanisms of catching up on lost sleep (sleep homeostasis), has previously focused on NREM sleep, where various circuits have been identified. That there is a special mechanism that also tracks time awake and compensates with REM sleep is intriguing.

In a broader context, the existence of REM rebound suggests that REM sleep must have a function, otherwise why catch up on it. There is a lot of literature that suggests REM contributes to emotional processing, for example. The new findings deepen our appreciation of REM regulation. As REM sleep is often disturbed in stress (e.g. post-traumatic stress disorder) and in depression, understanding more about REM regulation could ultimately aid treatments for people living with these conditions.

<https://doi.org/10.7554/eLife.92095.2.sa1>

Reviewer #2 (Public Review):

Maurer et al investigated the contribution of GAD2+ neurons in the preoptic area (POA), projecting to the tuberomammillary nucleus (TMN), to REM sleep regulation. They applied an elegant design to monitor and manipulate activity of this specific group of neurons: a GAD2-Cre mouse, injected with retrograde AAV constructs in the TMN, thereby presumably only targeting GAD2+ cells projecting to the TMN. Using this set-up in combination with technically challenging techniques including EEG with photometry and REM sleep deprivation, the authors found that this cell-type studied becomes active shortly (~40sec) prior to entering REM sleep and remains active during REM sleep. Moreover, optogenetic inhibition of GAD2+ cells inhibits REM sleep by a third, and also impairs the rebound in REM sleep in the following hour. Thus, the data makes a convincing case for a role of GAD2+ neurons in the POA projecting to the TMN in REM sleep regulation.

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Author Response

The following is the authors' response to the original reviews.

eLife assessment

This valuable study advances our understanding of the brain nuclei involved in rapid-eye movement (REM) sleep regulation. Using a combination of imaging, electrophysiology, and optogenetic tools, the study provides convincing evidence that inhibitory neurons in the preoptic area of the hypothalamus influence REM sleep. This work will be of interest to neurobiologists working on sleep and/or brain circuitry.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

This paper identifies GABA cells in the preoptic hypothalamus which are involved in REM sleep rebound (the increase in REM sleep) after selective REM sleep deprivation. By calcium photometry, these cells are most active during REM, and show more claim signals during REM deprivation, suggesting they respond to "REM pressure". Inhibiting these cells ontogenetically diminishes REM sleep. The optogenetic and photometry work

is carried out to a high standard, the paper is well-written, and the findings are interesting.

We thank the reviewer for the detailed feedback and thoughtful comments on how to improve our manuscript. To address the reviewer's concerns, we revised our discussion and added new data. Below, we address the concerns point by point.

Points that could be addressed or discussed:

(1) The circuit mechanism for REM rebound is not defined. How do the authors see REM rebound as working from the POAGAD2 cells? Although the POAGAD2 does project to the TMN, the actual REM rebound could be mediated by a projection of these cells elsewhere. This could be discussed.

We demonstrate that POA GAD2 → TMN cells become more frequently activated as the pressure for REMs builds up, whereas inhibiting these neurons during high REMs pressure leads to a suppression of the REMs rebound. It is not known how POA GAD2 → TMN cells encode increased REMs pressure and subsequently influence the REMs rebound. REMs deprivation was shown to change the intrinsic excitability of hippocampal neurons and impact synaptic plasticity (McDermott et al., 2003; Mallick and Singh, 2011; Zhou et al., 2020). We speculate that increased REMs pressure leads to an increase in the excitability of POA → TMN neurons, reflected in the increased number of calcium peaks. The increased excitability of POA GAD2 → TMN neurons in turn likely leads to stronger inhibition of downstream REM-off neurons. Consequently, as soon as REMs deprivation stops, there is an increased chance for entering REMs. The time course for how long it takes till the POA excitability resettles to its baseline consequently sets a permissive time window for increased amounts of REMs to recover its lost amount. For future studies, it would be interesting to map how quickly the excitability of POA neurons increases or decays as a function of the lost or recovered amount of REMs and unravel the cellular mechanisms underlying the elevated activity of POA GAD2 → TMN neurons during high REMs pressure, e.g., whether changes in the expression of ion channels contribute to increased excitability of these neurons (Donlea et al., 2014). As we mentioned in the Discussion, the POA also projects to other REMs regulatory brain regions such as the vPAG and LH. Therefore, it remains to be tested whether POA GAD2 → TMN neurons also innervate these brain regions to potentially regulate REMs homeostasis. We explicitly state this now in the revised Discussion.

(2) The "POAGAD2 to TMN" name for these cells is somewhat confusing. The authors chose this name because they approach the POAGAD2 cells via retrograde AAV labelling (rAAV injected into the TMN). However, the name also seems to imply that neurons (perhaps histamine neurons) in the TMN are involved in the REM rebound, but there is no evidence in the paper that this is the case. Although it is nice to see from the photometry studies that the histamine cells are selectively more active (as expected) in NREM sleep (Fig. S2), I could not logically see how this was a relevant finding to REM rebound or the subject of the paper. There are many other types of cells in the TMN area, not just histamine cells, so are the authors suggesting that these non-histamine cells in the TMN could be involved?

We acknowledge that other types of neurons in the TMN may also be involved in the REMs rebound, and therefore inhibition of histamine neurons by POA GAD2 → TMN neurons may not be the sole source of the observed effect. To stress that other neurons within the TMN and/or brain regions may also contribute to the REMs rebound, we have revised the Results section.

We performed complementary optogenetic inhibition experiments of TMN HIS neurons to investigate if suppression of these neurons is sufficient to promote REMs. We found that

SwiChR⁺⁺ mediated inhibition of TMN/HIS neurons increased the amount of REMs compared with recordings without laser stimulation in the same mice and eYFP mice with laser stimulation. Thus, while TMN/HIS neurons may not be the only downstream target of GABAergic POA neurons, these data suggest that they contribute to REMs regulation. We have incorporated these results in Fig. S4.

We further investigated whether the activity of TMN/HIS neurons changes between two REMs episodes. Assuming that REMs pressure inhibits the activity of REM-off histamine neurons, their firing rates should be highest right after REMs ends when REMs pressure is lowest, and progressively decay throughout the inter-REM interval, and reach their lowest activity right before the onset of REMs (Park et al., 2021), similar to the activity profile observed for vPAG REM-off neurons (Weber et al., 2018). We indeed found that TMN/HIS neurons display a gradual decrease in their activity throughout the inter-REM interval and thus potentially reflect the build up of REM pressure (Fig. S2F).

(3) It is a puzzle why most of the neurons in the POA seem to have their highest activity in REM, as also found by Miracca et al 2022, yet presumably some of these cells are going to be involved in NREM sleep as well. Could the same POA GAD2-TMN cells identified by the authors also be involved in inducing NREM sleep-inhibiting histamine neurons (Chung et al). And some of these POA cells will also be involved in NREM sleep homeostasis (e.g. Ma et al Curr Biol)? Is NREM sleep rebound necessary before getting REM sleep rebound? Indeed, can these two things (NREM and REM sleep rebound) be separated?

Previous studies have demonstrated that POA GABAergic neurons, including those projecting to the TMN, are involved in NREMs homeostasis (Sherin et al., 1998; Gong et al., 2004; Ma et al., 2019). Therefore, we predict that POA neurons that are involved in NREMs homeostasis are a subset of POA GAD2 → TMN neurons in our manuscript.

Using optrode recordings in the POA, we recently reported that 12.4% of neurons sampled have higher activity during NREMs compared with REMs; in contrast, 43.8% of neurons sampled have the highest activity during REMs compared with NREMs (Antila et al., 2022) indicating that the proportion of NREM max neurons is smaller compared with REM max neurons. These proportions of neurons are in agreement with previous results (Takahashi et al., 2009). Considering fiber photometry monitors the average activity of a population of neurons as opposed to individual neurons, it is possible that we recorded neural activity across heterogeneous populations and therefore our findings may disguise the neural activity of the low proportion of NREMs neurons. We previously reported the spiking activity of POA GAD2 → TMN neurons at the single cell level (Chung et al., 2017). We have noted in the manuscript that while the activity of POA GAD2 → TMN neurons is highest during REMs, the neural activity increases at NREMs → REMs transitions indicating these neurons also are active during NREMs.

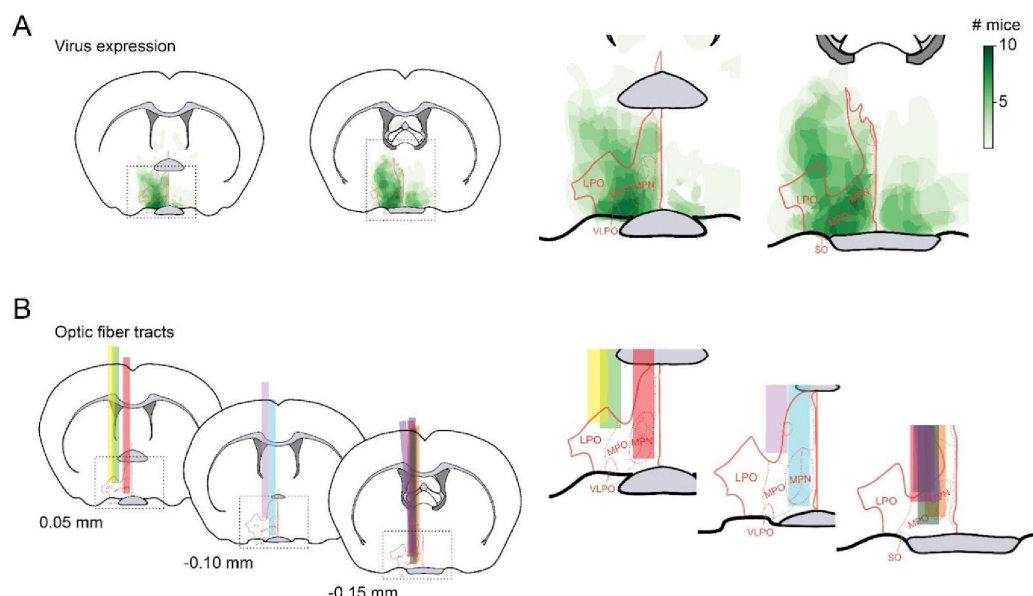
Using our REMs restriction protocol, we selectively restricted REMs leading to the subsequent rebound of REMs without affecting NREMs and consequently we did not find an increase in the amount of NREMs during the rebound or an increase in slow-wave activity, a key characteristic of sleep rebound that gradually dissipates during recovery sleep (Blake and Gerard, 1937; Williams et al., 1964; Rosa and Bonnet, 1985; Dijk et al., 1990; Neckelmann and Ursin, 1993; Ferrara et al., 1999). However, during total sleep deprivation when subjects are deprived of both NREMs and REMs, isolating NREMs and REMs rebound may not be attainable.

(4) Is it possible to narrow down the POA area where the GAD2 cells are located more precisely?

POA can be subdivided into anatomically distinct regions such as medial preoptic area, median preoptic area, ventrolateral preoptic area, and lateral preoptic area (MPO, MPN, VLPO, and LPO respectively). To quantify where the virus expressing GAD2 cells and optic fibers are located within the POA, we overlaid the POA coronal reference images (with red boundaries denoting these anatomically distinct regions) over the virus heat maps and optic fiber tracts from datasets used in Figure 1A. We found that virus expression and optic fiber tracts were located in the ventrolateral POA, lateral POA, and the lateral part of medial POA, and included this description in the text.

Author response image 1.

Location of virus expression (A) and optic fiber placement (B) within subregions of POA.



(5) It would be ideal to further characterize these particular GAD2 cells by RT-PCR or RNA seq. Which other markers do they express?

Single-cell RNA-sequencing of POA neurons has revealed an enormous level of molecular diversity, consisting of nearly 70 subpopulations based on gene expression of which 43 can be clustered into inhibitory neurons (Moffitt et al., 2018). One of the most studied subpopulation of POA sleep-active neurons contains the inhibitory neuropeptide galanin (Sherin et al., 1998; Gaus et al., 2002; Chung et al., 2017; Kroeger et al., 2018; Ma et al., 2019; Miracca et al., 2022). Galanin neurons have been demonstrated to innervate the TMN (Sherin et al., 1998) yet, within the galanin neurons 7 distinct clusters exist based on unique gene expression (Moffitt et al., 2018). In addition to galanin, we have previously performed single-cell RNA-seq on POA GAD2 → TMN neurons and identified additional neuropeptides such as cholecystikinin (CCK), corticotropin-releasing hormone (CRH), prodynorphin (PDYN), and tachykinin 1 (TAC1) as subpopulations of GABAergic POA sleep-active neurons (Chung et al., 2017; Smith et al., 2023). Like galanin, these neuropeptides can also be divided into multiple subtypes as well (Chen et al., 2017; Moffitt et al., 2018). Thus while these molecular markers for POA neurons are immensely diverse, we agree that characterizing the molecular identity of POA GAD2 → TMN neurons and investigating the functional relevance of these neuropeptides in the context of REMs homeostasis would enrich our understanding of a neural circuit involved in REMs homeostasis and can stand as a separate extension of this manuscript.

Reviewer #2 (Public Review):

Maurer et al investigated the contribution of GAD2+ neurons in the preoptic area (POA), projecting to the tuberomammillary nucleus (TMN), to REM sleep regulation. They applied an elegant design to monitor and manipulate the activity of this specific group of neurons: a GAD2-Cre mouse, injected with retrograde AAV constructs in the TMN, thereby presumably only targeting GAD2+ cells projecting to the TMN. Using this set-up in combination with technically challenging techniques including EEG with photometry and REM sleep deprivation, the authors found that this cell-type studied becomes active shortly (≈ 40 sec) prior to entering REM sleep and remains active during REM sleep. Moreover, optogenetic inhibition of GAD2+ cells inhibits REM sleep by a third and also impairs the rebound in REM sleep in the following hour. Despite a few reservations or details that would benefit from further clarification (outlined below), the data makes a convincing case for the role of GAD2+ neurons in the POA projecting to the TMN in REM sleep regulation.

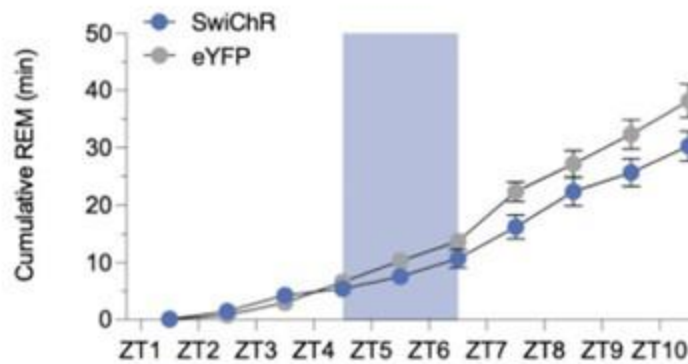
We thank the reviewer for the thorough assessment of our study and supportive comments. We have addressed your concerns in the revised manuscript, and our point by point response is provided below.

The authors found that optogenetic inhibition of GAD2+ cells suppressed REM sleep in the hour following the inhibition (e.g. Fig2 and Fig4). If the authors have the data available, it would be important to include the subsequent hours in the rebound time (e.g. from ZT8.5 to ZT24) to test whether REM sleep rebound remains impaired, or recovers, albeit with a delay.

We thank the reviewer for this comment and agree that it would be interesting to know how REMs changes for a longer period of time throughout the rebound phase. For Fig. 2, we did not record the subsequent hours. For Fig 4, we recorded the subsequent rebound between ZT7.5 and 10.5. When we compare the REMs amount during this 4 hr interval, the SwiChR mice have less REMs compared with eYFP mice with marginal significance (unpaired t-test, $p=0.0641$). We also plotted the cumulative REMs amount during restriction and rebound phases, and found that the cumulative amount of REMs was still lower in SwiChR mice than eYFP mice at ZT 10.5 (Author response image 2). Therefore, it will be interesting to record for a longer period of time to test when the SwiChR mice compensate for all the REMs that was lost during the restriction period.

Author response image 2.

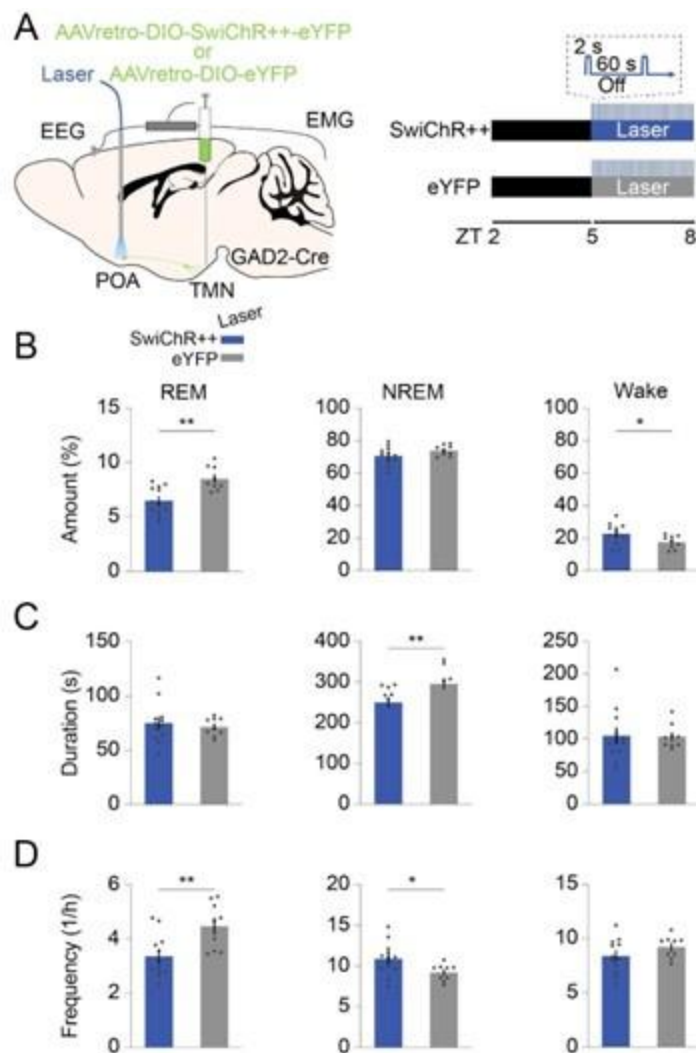
Cumulative amount of REMs during REMs deprivation and rebound combined with optogenetic stimulation in eYFP and SwiChR groups. This data is shown as bar graphs in Figure 4.



REM sleep is under tight circadian control (e.g. Wurts et al., 2000 in rats; Dijk, Czeisler 1995 in humans). To contextualize the results, it would be important to mention that it is not clear if the role of the manipulated neurons in REM sleep regulation hold at other circadian times of the day.

Author response image 3.

Inhibiting POA GAD2 → TMN neurons at ZT5-8 reduces REMs. (A) Schematic of optogenetic inhibition experiments. (B) Percentage of time spent in REMs, NREMs and wakefulness with laser in SwiChR⁺⁺ and eYFP mice. Unpaired t-tests, $p = 0.0013$, 0.0469 for REMs and wakeamount. (C) Duration of REMs, NREMs, and wake episodes. Unpaired t-tests, $p = 0.0113$ for NREMs duration. (D) Frequency of REMs, NREMs, and wake episodes. Unpaired t-tests, $p = 0.0063$, 0.0382 for REMs and NREMs frequency.



REMs propensity is largest towards the end of the light phase (Czeisler et al., 1980; Dijk and Czeisler, 1995; Wurts and Edgar, 2000). As a control, we therefore performed the optogenetic inhibition experiments of POA GAD2 → TMN neurons during ZT5-8 (Author response image 3). Similar to our results in Figure 2, we found that SwiChR-mediated inhibition of POA GAD2 → TMN neurons attenuated REMs compared with eYFP laser sessions. These findings suggest our results are consistent at other circadian times of the day.

The effect size of the REM sleep deprivation using the vibrating motor method is unclear. In FigS4-D, the experimental mice reduce their REM sleep to 3% whereas the control mice spend 6% in REM sleep. In Fig4, mice are either subjected to REM sleep deprivation with the vibrating motor (controls), or REM sleep deprivations + optogenetics (experimental mice).

The control mice (vibrating motor) in Fig4 spend 6% of their time in REM sleep, which is double the amount of REM sleep compared to the mice receiving the same treatment in FigS4-D. Can the authors clarify the origin of this difference in the text?

The effect size for REM sleep deprivation is now added in the text.

It is important to note that these figures are analyzing two different intervals of the REMs restriction. In Fig. S4D, we analyzed the total amount of REMs over the entire 6 hr restriction

interval (ZT1.5-7.5). In Fig. 4, we analyzed the amount of REMs only during the last 3 hr of restriction (ZT4.5-7.5) as optogenetic inhibition was performed only during the last 3 hrs when the REMs pressure is high. In Fig. S4D, we looked at the amount of REMs during ZT1.5-4.5 and 4.5-7.5 and found that the amount of REMs during ZT4.5-7.5 ($4.46 \pm 0.25\%$; mean $\pm$ s.e.m.) is indeed higher than ZT 1.5-4.5 ($1.66 \pm 0.62\%$), and is comparable to the amount of REMs during ZT4.5-7.5 in eYFP mice ($5.95 \pm 0.52\%$) in Fig. 4. We now clearly state in the manuscript at which time points we analyzed the amount, duration and frequency of REMs.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

(1) A few further citations suggested: Discussion "The TMN contains histamine producing neurons and antagonizing histamine neurons causes sleepiness..." It would be appropriate to cite Uygun DS et al 2016 J Neurosci (PMID: 27807161) here. Using the same HDC-Cre mice as used by Maurer et al., Uygun et al found that selectively increasing GABAergic inhibition onto histamine neurons produced NREM sleep.

We apologize for omitting this important paper. In the revised manuscript, we added this citation.

(2) Materials and Methods.

Although the JAX numbers are given for the mouse lines based on researchers generously donating to JAX for others to use, please cite the papers corresponding to the GAD2-ires-Cre and HDC-ires-Cre mouse lines deposited at JAX.

GAD2-ires-Cre was described in Taniguchi H et al., 2011, Neuron (PMID: 21943598).

The construction of the HDC-ires-CRE line is described in Zecharia AY et al J Neurosci et al 2012 (PMID: 22993424).

We have now added these important citations in the revised manuscript.

(3) Similarly, for the viruses, please provide the citations for the AAV constructs that were donated to Addgene.

We have now added these citations in the revised manuscript.

Reviewer #2 (Recommendations For The Authors):

The authors rely heavily on their conclusions by using an optogenetic tool that inhibits the activity of GAD2+ neurons, however, it is not shown that these neurons are indeed inhibited as expected. An alternative approach to tackle this could be the application of a different technique to achieve the same output (e.g. chemogenetics). However, both experiments (confirmation of inhibition, or using a different technique) would require a significant amount of work, and given the numerous studies out there showing that these optogenetic tools tend to work, may not be necessary. Hence the authors could also cite a similar study that used a likewise construct and where it was indeed shown that this technique works (i.e. similar retrograde optogenetic construct with Cre dependent expression combined with electrophysiological recordings).

This laser stimulation protocol was designed based on previous reports of sustained inhibition using the same inhibitory opsin and our prior results that recapitulate similar findings as inhibitory chemogenetic techniques (Iyer et al., 2016; Kim et al., 2016; Wiegert et al., 2017; Stucynski et al., 2022). We have now added this description in the Result section.

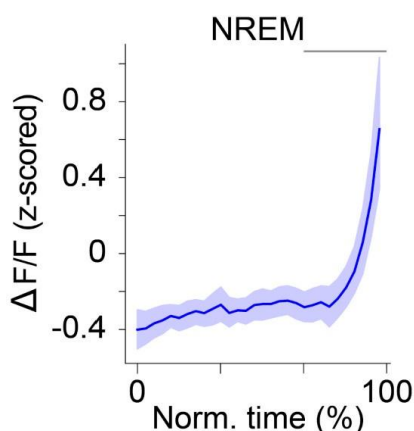
Fig1A - Right: the virus expression graphs are great and give a helpful insight into the variability. The image on the left (GCAMP+ cells) is less clear, the GCAMP+ cells don't differentiate well from the background. Perhaps the whole brain image with inset in POA can show the GCAMP expression more convincingly.

We have added a histology picture showing the whole brain image with inset in the POA in the updated Fig. 1A .

Statistics: The table is very helpful. Based on the degrees of freedom, it seems that in some instances the stats are run on the recordings rather than on the individual mice (e.g. Fig1). It could be considered to use a mixed model where subjects as taken into account as a factor.

Author response image 4.

$\Delta F/F$ activity of POA GAD2 $\rightarrow$ TMN neurons during NREMs. The duration of NREMs episodes was normalized in time, ranging from 0 to 100%. Shading, $\pm$ s.e.m. Pairwise t-tests with Holm-Bonferroni correction $p = 5.34 \times 10^{-4}$ between 80 and 100. Graybar, intervals where $\Delta F/F$ activity was significantly different from baseline (0 to 20%, the first time bin). $n = 10$ mice. In Fig. 1E , we ran stats based on the recordings. In this data set, we ran stats based on the individual mice, and found that the activity also gradually increased throughout NREMs episodes.



There is an effect of laser in Fig2 on REM sleep amount, as well as an interaction effect with virus injection (from the table). Therefore, it would be helpful for the reader to also show REM sleep data from the control group (laser stimulation but no active optogenetics construct) in Fig 2.

To properly control laser and virus effect, we performed the same laser stimulation experiments in eYFP control mice (expressing only eYFP without optogenetic construct, SwiChR++) and the data is provided in Fig 2C .

Fig3B: At the start of the rebound of REM sleep, there is a massive amount of wakefulness, also reflected in the change of spectral composition. Could you comment on the text about what is happening here?

We quantified the amount of wakefulness during the first hour of REMs rebound and found that indeed there is no significant difference in wakefulness between REM restriction and baseline control conditions (Fig. S4H). Therefore, while the representative image in Fig 3B

shows increased wakefulness at the beginning of REMs rebound, we do not think the overall amount of wakefulness is increased.

Fig 4, supplementary data: it would be helpful for the reader to have mentioned in the text the effect size of the REM sleep restriction protocol (e.g. mean and standard deviation).

Thank you for this suggestion. We have now added the effect size for the REM sleep restriction experiments in the main text.

REM sleep restriction and photometry experiment: could be improved by adding within the main body of text that, in order to conduct the photometry experiment in the last hours of REM sleep deprivation, the manual REM sleep deprivation had to be applied, because the vibrating motor technique disturbed the photometry recordings.

Thank you for this suggestion. We have added the description in the main text.

Suggestion to build further on the already existing data (not for this paper): you have a powerful dataset to test whether REM sleep pressure builds up during wakefulness or NREM sleep, by correlating when your optogenetic treatment occurs (NREM or wakefulness), with the subsequent rebound in REM sleep (see also Endo et al., 1998; Benington and Heller, 1994; Franken 2001).

We thank the reviewer for this excellent suggestion. We plan to carry out this experiment in the future.

References

- Antila, H., Kwak, I., Choi, A., Pisciotto, A., Covarrubias, I., Baik, J., et al. (2022). A noradrenergic-hypothalamic neural substrate for stress-induced sleep disturbances. *Proc. Natl. Acad. Sci.* 119, e2123528119. doi: 10.1073/pnas.2123528119.
- Blake, H., and Gerard, R. W. (1937). Brain potentials during sleep. *Am. J. Physiol.-Leg. Content* 119, 692–703. doi: 10.1152/ajplegacy.1937.119.4.692.
- Chen, R., Wu, X., Jiang, L., and Zhang, Y. (2017). Single-Cell RNA-Seq Reveals Hypothalamic Cell Diversity. *Cell Rep.* 18, 3227–3241. doi: 10.1016/j.celrep.2017.03.004.
- Chung, S., Weber, F., Zhong, P., Tan, C. L., Nguyen, T., Beier, K. T., et al. (2017). Identification of Preoptic Sleep Neurons Using Retrograde Labeling and Gene Profiling. *Nature* 545, 477–481. doi: 10.1038/nature22350.
- Czeisler, C. A., Zimmerman, J. C., Ronda, J. M., Moore-Ede, M. C., and Weitzman, E. D. (1980). Timing of REM sleep is coupled to the circadian rhythm of body temperature in man. *Sleep* 2, 329–346.
- Dijk, D. J., Brunner, D. P., Beersma, D. G., and Borbély, A. A. (1990). Electroencephalogram power density and slow wave sleep as a function of prior waking and circadian phase. *Sleep* 13, 430–440. doi: 10.1093/sleep/13.5.430.
- Dijk, D. J., and Czeisler, C. A. (1995). Contribution of the circadian pacemaker and the sleep homeostat to sleep propensity, sleep structure, electroencephalographic slow waves, and sleep spindle activity in humans. *J. Neurosci. Off. J. Soc. Neurosci.* 15, 3526–3538. doi: 10.1523/JNEUROSCI.15-05-03526.1995.
- Donlea, J. M., Pimentel, D., and Miesenböck, G. (2014). Neuronal machinery of sleep homeostasis in *Drosophila*. *Neuron* 81, 860–872. doi: 10.1016/j.neuron.2013.12.013.

- Ferrara, M., De Gennaro, L., Casagrande, M., and Bertini, M. (1999). Auditory arousal thresholds after selective slow-wave sleep deprivation. *Clin. Neurophysiol. Off. J. Int. Fed. Clin. Neurophysiol.* 110, 2148–2152. doi: 10.1016/s1388-2457(99)00171-6.
- Gaus, S. E., Strecker, R. E., Tate, B. A., Parker, R. A., and Saper, C. B. (2002). Ventrolateral preoptic nucleus contains sleep-active, galaninergic neurons in multiple mammalian species. *Neuroscience* 115, 285–294. doi: 10.1016/S0306-4522(02)00308-1.
- Gong, H., McGinty, D., Guzman-Marin, R., Chew, K.-T., Stewart, D., and Szymusiak, R. (2004). Activation of c-fos in GABAergic neurones in the preoptic area during sleep and in response to sleep deprivation. *J. Physiol.* 556, 935–946. doi: 10.1113/jphysiol.2003.056622.
- Iyer, S. M., Vesuna, S., Ramakrishnan, C., Huynh, K., Young, S., Berndt, A., et al. (2016). Optogenetic and chemogenetic strategies for sustained inhibition of pain. *Sci. Rep.* 6, 30570. doi: 10.1038/srep30570.
- Kim, H., Ährlund-Richter, S., Wang, X., Deisseroth, K., and Carlén, M. (2016). Prefrontal Parvalbumin Neurons in Control of Attention. *Cell* 164, 208–218. doi: 10.1016/j.cell.2015.11.038.
- Kroeger, D., Absi, G., Gagliardi, C., Bandaru, S. S., Madara, J. C., Ferrari, L. L., et al. (2018). Galanin neurons in the ventrolateral preoptic area promote sleep and heat loss in mice. *Nat. Commun.* 9, 4129. doi: 10.1038/s41467-018-06590-7.
- Ma, Y., Miracca, G., Yu, X., Harding, E. C., Miao, A., Yustos, R., et al. (2019). Galanin Neurons Unite Sleep Homeostasis and α 2-Adrenergic Sedation. *Curr. Biol. CB* 29, 3315–3322.e3. doi: 10.1016/j.cub.2019.07.087.
- Mallick, B. N., and Singh, A. (2011). REM sleep loss increases brain excitability: role of noradrenaline and its mechanism of action. *Sleep Med. Rev.* 15, 165–178. doi: 10.1016/j.smrv.2010.11.001.
- McDermott, C. M., LaHoste, G. J., Chen, C., Musto, A., Bazan, N. G., and Magee, J. C. (2003). Sleep deprivation causes behavioral, synaptic, and membrane excitability alterations in hippocampal neurons. *J. Neurosci. Off. J. Soc. Neurosci.* 23, 9687–9695. doi: 10.1523/JNEUROSCI.23-29-09687.2003.
- Miracca, G., Anuncibay-Soto, B., Tossell, K., Yustos, R., Vyssotski, A. L., Franks, N. P., et al. (2022). NMDA Receptors in the Lateral Preoptic Hypothalamus Are Essential for Sustaining NREM and REM Sleep. *J. Neurosci.* 42, 5389–5409. doi: 10.1523/JNEUROSCI.0350-21.2022.
- Moffitt, J. R., Bambah-Mukku, D., Eichhorn, S. W., Vaughn, E., Shekhar, K., Perez, J. D., et al. (2018). Molecular, spatial, and functional single-cell profiling of the hypothalamic preoptic region. *Science* 362. doi: 10.1126/science.aau5324.
- Neckelmann, D., and Ursin, R. (1993). Sleep stages and EEG power spectrum in relation to acoustical stimulus arousal threshold in the rat. *Sleep* 16, 467–477.
- Park, S.-H., Baik, J., Hong, J., Antila, H., Kurland, B., Chung, S., et al. (2021). A probabilistic model for the ultradian timing of REM sleep in mice. *PLOS Comput. Biol.* 17, e1009316. doi: 10.1371/journal.pcbi.1009316.
- Rosa, R. R., and Bonnet, M. H. (1985). Sleep stages, auditory arousal threshold, and body temperature as predictors of behavior upon awakening. *Int. J. Neurosci.* 27, 73–83. doi: 10.3109/00207458509149136.
- Sherin, J. E., Elmquist, J. K., Torrealba, F., and Saper, C. B. (1998). Innervation of histaminergic tuberomammillary neurons by GABAergic and galaninergic neurons in the ventrolateral

preoptic nucleus of the rat. *J. Neurosci. Off. J. Soc. Neurosci.* 18, 4705–4721.

Smith, J., Honig-Frand, A., Antila, H., Choi, A., Kim, H., Beier, K. T., et al. (2023). Regulation of stress-induced sleep fragmentation by preoptic glutamatergic neurons. *Curr. Biol. CB*, S0960-9822(23)01585–3. doi: 10.1016/j.cub.2023.11.035.

Stucynski, J. A., Schott, A. L., Baik, J., Chung, S., and Weber, F. (2022). Regulation of REM sleep by inhibitory neurons in the dorsomedial medulla. *Curr. Biol. CB* 32, 37-50.e6. doi: 10.1016/j.cub.2021.10.030.

Takahashi, K., Lin, J.-S., and Sakai, K. (2009). Characterization and mapping of sleep-waking specific neurons in the basal forebrain and preoptic hypothalamus in mice. *Neuroscience* 161, 269–292. doi: 10.1016/j.neuroscience.2009.02.075.

Weber, F., Hoang Do, J. P., Chung, S., Beier, K. T., Bikov, M., Saffari Doost, M., et al. (2018). Regulation of REM and Non-REM sleep by periaqueductal GABAergic neurons. *Nat. Commun.* 9, 1–13. doi: 10.1038/s41467-017-02765-w.

Wiegert, J. S., Mahn, M., Prigge, M., Printz, Y., and Yizhar, O. (2017). Silencing Neurons: Tools, Applications, and Experimental Constraints. *Neuron* 95, 504–529. doi: 10.1016/j.neuron.2017.06.050.

Williams, H. L., Hammack, J. T., Daly, R. L., Dement, W. C., and Lubin, A. (1964). RESPONSES TO AUDITORY STIMULATION, SLEEP LOSS AND THE EEG STAGES OF SLEEP. *Electroencephalogr. Clin. Neurophysiol.* 16, 269–279. doi: 10.1016/0013-4694(64)90109-9.

Wurts, S. W., and Edgar, D. M. (2000). Circadian and homeostatic control of rapid eye movement (REM) sleep: promotion of REM tendency by the suprachiasmatic nucleus. *J. Neurosci. Off. J. Soc. Neurosci.* 20, 4300–4310. doi: 10.1523/JNEUROSCI.20-11-04300.2000.

Zhou, Y., Lai, C. S. W., Bai, Y., Li, W., Zhao, R., Yang, G., et al. (2020). REM sleep promotes experience-dependent dendritic spine elimination in the mouse cortex. *Nat. Commun.* 11, 4819. doi: 10.1038/s41467-020-18592-5.